Unique Equilibrium in Models with Mixed Logit Demand Under Variance Constraints*

Diego Cussen†

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Abstract

I prove the existence and uniqueness of a Bertrand-Nash equilibrium in a mixed logit demand model with linear utility and multiproduct firms. The proof relies on a uniform convergence result showing that mixed logit profit functions, and their derivatives, converge to their logit counterparts as the variances of random coefficients shrink, combined with an existing uniqueness result for logit demand. This approach requires imposing variance constraints, which may bind in some applications. I propose a diagnostic for checking whether the constraints bind for a given model and parameter estimates.

Keywords: Mixed logit, Bertrand-Nash equilibrium, uniqueness.

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†Department of Economics, New York University (email: dc5004@nyu.edu).

1 Introduction

I prove the existence of a unique Bertrand-Nash equilibrium in pure strategies for a subset of models with (1) mixed logit demand systems, (2) a continuum of consumer types, (3) linear utility, (4) constant returns to scale, (5) multiproduct firms, and (6) arbitrary product heterogeneity (Boyd and Mellman, 1980; Cardell and Dunbar, 1980; Berry et al., 1995). This is the workhorse model of demand estimation (Conlon and Gortmaker, 2020), and these existence and uniqueness proofs are, to the best of my knowledge, the first of their kind for mixed logit models. The core idea for the proofs is that mixed multinomial logit (henceforth mixed logit) market shares, and their partial derivatives with respect to prices, converge *uniformly* to their multinomial logit (henceforth logit) counterparts as the variances of random coefficients go to zero.

The idea is intuitive but the proof is not straightforward. With this result in hand, I use the uniqueness of Bertrand-Nash equilibrium in logit models (Konovalov and Sándor, 2010), and Morrow and Skerlos (2010a)'s characterization of equilibrium conditions in mixed logit, to complete the proof of existence and uniqueness of equilibrium in mixed logit.

The drawback of this method of proof is that it requires restrictions on the variances of the random coefficients. I explore the extent to which these constraints are binding, and how to generalize the results to a wider class of models.

The results of this paper provide information about model primitives for mixed logit models that yield unique equilibria, and matter for identification, policy evaluation, and optimal instruments computation (Jovanovic, 1989; Berry et al., 1995; Nocke and Schutz, 2018).

Related Literature. Aksoy-Pierson et al. (2013) prove the existence and uniqueness of a Bertrand-Nash equilibrium under mixed logit demand with single product firms, discrete consumer types, and market share bounds. Nocke and Schutz (2025) prove existence results using potential games for a large class of oligopoly models with flexible substitution patterns, but excluding mixed logit.

Existence of a unique equilibrium also matters in dynamic games where a stage of game includes oligopoly under mixed logit demand. In such a case, existence and uniqueness are useful for constructing the payoff function of the dynamic game. Examples are two-stage bargaining games, like Crawford and Yurukoglu (2012). Existence of equilibrium in the oligopoly competition game also matters when bargaining and competition are simultaneous, as in Crawford et al. (2018). This requirement also extends to models with infinite horizon (Bajari et al., 2007)

This paper is organized as follows. Section 2 sets out primitives. Section 3 proves the mixed logit demand models we consider converge uniformly to logit demand as the variances of the random coefficients go to zero. Section 4 presents the existence and global and local uniqueness results. Section 5 proposes a method to explore the extent to which the variance constraints are binding. Section 6 shows how to relax my assumptions. Section 7 concludes.

2 The Model

This section states the primitives of the mixed logit models we consider, states the assumptions I use in the proofs, and presents a continuous differentiability result for firms' profit functions. The notation is based on, but different from, that in Morrow and Skerlos (2010a).

2.1 Primitives for Mixed Logit

There are $j \in \{1, ..., J\} \equiv \mathcal{J}$ firms. The set of differentiated products each firm sells is $l \in \mathcal{L}_j$, with $|\mathcal{L}_j| = L_j$. Firms sell mutually exclusive sets of products, and the set of all $L \equiv \sum_j L_j$ products is $\mathcal{L}$. The set of products sold by firms other than j is $\mathcal{L}_{-j}$, with $|\mathcal{L}_{-j}| = L_{-j}$. The function $j(l)$ maps product l to the firm that sells it.

Each product is sold at price p_l. The vector of L_j prices firm j sets is $\mathbf{p}_j \subset \mathbb{R}_+^{L_j}$, where $\mathbb{R}_+$ are the non-negative real numbers and **bold** denotes a vector. The vector of prices set by all other firms is $\mathbf{p}_{-j} \in \mathbb{R}_+^{L_{-j}}$. The vector of all prices is $\mathbf{p} \in P = \mathbb{R}_+^L$. Firms compete in prices. Consumers can also choose an outside good, whose price we normalize to zero. The equilibrium concept is Bertrand-Nash.

Firms have constant marginal costs. The marginal cost of good l is $c_l > 0$. The vector of all marginal costs of j is $\boldsymbol{c}_j \in \mathbb{R}_{++}^{L_j}$, where $\mathbb{R}_{++}$ denotes the positive real numbers. The vector of all marginal costs in the model is $\boldsymbol{c} \in \mathbb{R}_{++}^L$.

Each product has a vector $\mathbf{x}_l \in X \subset \mathbb{R}^{\dim(X)}$ of non-price observable characteristics. $\mathbf{x}_l$ may also be a polynomial of the values of characteristics. For all $l \in \{0\} \cup \mathcal{L}$, all unobservable characteristics are summarized in $\xi_l \in \Xi$. All $\mathbf{x}_l$ and ξ_l are exogenous. Note that ξ_l may include a product fixed effect. Therefore, since characteristics are exogenous, we may reinterpret ξ_l as summarizing both unobserved characteristics and any observable characteristics whose coefficients are always constant.

Without loss, consumers i lie in a continuum of Lebesgue measure one. Each consumer is characterized by a vector of coefficients $\boldsymbol{\theta}_i \equiv (\alpha_i, \boldsymbol{\beta}_i) \in \mathbb{R}_{++} \times \mathbb{R}^{\dim(X)} \equiv \Theta$. α_i is i's price sensitivity (or marginal utility of income), while $\boldsymbol{\beta}_i$ are the coefficients attached to observable characteristics $\mathbf{x}_l$. α_i has a shifted log-normal distribution, i.e., there exists $\underline{\alpha} \in \mathbb{R}_+$ such that $\ln(\alpha_i - \underline{\alpha}) \sim N(\mu_\alpha, \sigma_\alpha^2)$.

$\boldsymbol{\beta}_i$ is jointly normally distributed, with expectation $\overline{\boldsymbol{\beta}}$ and variance-covariance matrix $\Sigma_{\boldsymbol{\beta}}$. The main analysis assumes that $(\alpha_i - \underline{\alpha})$ and $\boldsymbol{\beta}_i$ are independent: $(\alpha_i - \underline{\alpha}) \perp \boldsymbol{\beta}_i$. Let $\widetilde{\boldsymbol{\theta}}_i \equiv (\ln(\alpha_i - \underline{\alpha}), \boldsymbol{\beta}_i)$, so there is a bijection between $\boldsymbol{\theta}_i$ and $\widetilde{\boldsymbol{\theta}}_i$. Let $\mu \equiv (\mu_\alpha, \underline{\alpha}, \overline{\boldsymbol{\beta}})$. Given μ, the joint distribution of $\widetilde{\boldsymbol{\theta}}_i$ is given by the variance-covariance matrix Σ, which has the form

$$\Sigma = \begin{bmatrix} \sigma_\alpha^2 & \mathbf{0}_K^T \\ \mathbf{0}_K & \Sigma_{\boldsymbol{\beta}} \end{bmatrix}$$

where $\mathbf{0}_K$ is a $K-$element column vector of zeros and T indicates transpose. We denote the probability measure of $\widetilde{\boldsymbol{\theta}}_i$ with ρ, which is parameterized by μ and Σ. It will be useful to specify a metric space for the variance-covariance matrices. To that end, consider the set of positive semi-definite, symmetric matrices with zero covariance between α_i and $\boldsymbol{\beta}_i$:

$$S = \left\{ A \in \mathbb{R}^{\dim(\Theta) \times \dim(\Theta)} : A \text{ is a variance-covariance matrix, } A_{1,h} = 0 \, \forall h > 1 \right\}.$$

If A is a variance-covariance matrix then it is positive semi-definite ($A \succeq 0$) and symmetric ($A = A^T$). The metric space we will use is $(S, \|\cdot\|_\infty)$,where $\|\cdot\|_\infty$ is the entry-wise maximum norm. Note that $\mathbf{0}_\Sigma \in S$, where $\mathbf{0}_\Sigma$ is a $\dim(\Theta) \times \dim(\Theta)$ matrix of zeros.

Consumer i's indirect utility from choosing $l \in \mathcal{L}$ is

$$U_{il}(p_l, \boldsymbol{\theta}_i, \epsilon_{il}) = u_l(p_l, \boldsymbol{\theta}_i) + \epsilon_{il}$$

where ϵ_{il} is a shock with type 1 extreme value distribution i.i.d across products and consumers.

The following assumptions summarize the specification of indirect utilities and the distribution of random coefficients. In Section 6 I explain how these assumptions may be relaxed/generalized, at the cost of no longer characterizing the models to which my results apply in terms of primitives.

Assumption 2.1 (Linear Utility Mixed Logit). *For each consumer i and product l, systematic indirect utility satisfies*

$$u_l(p_l, \boldsymbol{\theta}_i) = -\alpha_i p_l + \mathbf{x}_l^\top \boldsymbol{\beta}_i + \xi_l, \quad \forall l \in \mathcal{L}. \tag{1}$$

The outside-good utility is

$$u_0 = \xi_0.$$

Assumption 2.1 excludes the presence of random coefficients in the indirect utility of the outside good. This is because while such random coefficients may improve the fit of a model to the data, they are not relevant for our proof, and their exclusion reduces notational clutter.

Assumption 2.2 establishes that price sensitivities must have a lower bound away from zero.

Assumption 2.2 (Minimum Price Sensitivity). *For all i, $\alpha_i \geq \underline{\alpha} > 0$.*

Assumption 2.2, is consistent with downward sloping demand functions, positive marginal utility of income, and the intuition that existence of equilibrium in mixed logit models requires that price sensitivities be sufficiently positive.[1] Assumption 2.2 is not consistent, however, with a normal distribution of price sensitivities—a distribution sometimes used in applied work. Note that if all consumers have the same price sensitivity, a negative-sloping demand function implies Assumption 2.2 mechanically.

Assumption 2.3 (Normal-Log-normal Random Coefficients). *The joint distribution of $\widetilde{\boldsymbol{\theta}}_i$ follows:*

1. $(\alpha_i - \underline{\alpha}) \sim LogNormal(\mu_\alpha, \sigma_\alpha^2)$.

2. $\boldsymbol{\beta}_i \sim N(\overline{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}})$.

3. $(\alpha_i - \underline{\alpha}) \perp \boldsymbol{\beta}_i$.

Assumptions 2.2 and 2.3 mean α_i has a shifted log-normal distribution. This can be made arbitrarily close to a normal distribution, however, because Assumptions 2.2 is consistent with and arbitrarily small $\underline{\alpha}$ (see Section 6.2). The log-normal distribution has the attractive property that it is consistent with linear-in-price utility being a first order approximation of log residual income utility with a log-normally distributed income (Berry et al., 1999).

[1] I thank Chris Conlon for a discussion of this point.

2.2 Logit as a Special Case

Let $\overline{\alpha} \equiv \exp(\mu_\alpha) + \underline{\alpha}$. For every mixed logit model with distribution parameters μ_α, $\overline{\boldsymbol{\beta}}$, $\underline{\alpha}$, and variance covariance matrix Σ there exists a logit model with linear utility that satisfies Assumption 2.4:

Assumption 2.4 (Linear Utility Logit)**.** *For each consumer i and product l, systematic indirect utility satisfies*

$$u_l(p_l) = -\overline{\alpha} p_l + \mathbf{x}_l^T \overline{\boldsymbol{\beta}} + \xi_l \qquad \forall l \in \mathcal{L} \tag{2}$$

and

$$u_0 = \xi_0.$$

2.3 Market Shares, Profits, and Partial Derivatives

I close Section 2 with a presentation of expressions for market shares, their partial derivatives, and a proof that market shares and profits are twice continuously differentiable with respect to prices. Given $\boldsymbol{\theta}_i$, the logit choice probability of choosing good $l \in \mathcal{L} \cup \{0\}$ is

$$s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) = \frac{e^{u_l(p_l, \boldsymbol{\theta}_i)}}{e^{u_0(\boldsymbol{\theta}_i)} + \sum_{k \in \mathcal{L}} e^{u_k(p_k, \boldsymbol{\theta}_i)}}$$

so the market shares are

$$s_l(\mathbf{p}; \mu, \Sigma) = \int \frac{e^{u_l(p_l, \boldsymbol{\theta}_i)}}{e^{u_0(\boldsymbol{\theta}_i)} + \sum_{k \in \mathcal{L}} e^{u_k(p_k, \boldsymbol{\theta}_i)}} d\rho(\widetilde{\boldsymbol{\theta}}_i; \mu, \Sigma) \tag{3}$$

Ignoring fixed costs without loss, firm j's expected profits function is

$$\pi_j(\mathbf{p}_j, \mathbf{p}_{-j}; \mu, \Sigma) = \sum_{l \in \mathcal{L}_j} (p_l - c_l) s_l(\mathbf{p}; \mu, \Sigma). \tag{4}$$

Using matrix notation, Morrow and Skerlos (2010a) show that the matrix of first partial derivatives of market shares with respect to prices is

$$\frac{\partial \mathbf{s}(\mathbf{p}; \mu, \Sigma)}{\partial \mathbf{p}} = \Lambda(\mathbf{p}; \mu, \Sigma) - \Gamma(\mathbf{p}; \mu, \Sigma)$$

where $\Lambda(\mathbf{p}; \mu, \Sigma)$ is a diagonal $L \times L$ matrix with entries

$$\Lambda_{ll} = \int -\alpha_i s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) d\rho(\widetilde{\boldsymbol{\theta}}_i; \mu, \Sigma) \tag{5}$$

and $\Gamma(\mathbf{p}; \mu, \Sigma)$ is a dense matrix of entries

$$\Gamma_{kl} = \int -\alpha_i s_k^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) d\rho(\widetilde{\boldsymbol{\theta}}_i; \mu, \Sigma). \tag{6}$$

$\Gamma(\mathbf{p}; \mu, \Sigma)$ is symmetric if utility is linear in price (Assumption 2.1). Morrow and Skerlos (2010b) present analogous versions of $\Lambda(\mathbf{p}; \mu, \Sigma)$ and $\Gamma(\mathbf{p}; \mu, \Sigma)$ for logit.

For each j, the $L_j \times L_j$ matrices $\Lambda_j(\mathbf{p};\mu,\Sigma)$ and $\Gamma_j(\mathbf{p};\mu,\Sigma)$ contain the entries in of $\Lambda_j(\mathbf{p};\mu,\Sigma)$ and $\Gamma_j(\mathbf{p};\mu,\Sigma)$ in their rows and columns $L_{j-1}+1$ through $L_{j-1}+L_j$ (with $L_0 = 0$). Thus, letting $D_{\mathbf{y}}h$ denote the array of partial derivatives of h with respect to $\mathbf{y}$, we can write j's first-order condition as

$$D_{\mathbf{p}_j}\pi_j(\mathbf{p}_j,\mathbf{p}_{-j};\mu,\Sigma) = \mathbf{s}_j(\mathbf{p};\mu,\Sigma) + [\Lambda_j(\mathbf{p};\mu,\Sigma) - \Gamma_j(\mathbf{p};\mu,\Sigma)](\mathbf{p}_j - \boldsymbol{c}_j) = 0.$$

Stacking the J Jacobians $D_{\mathbf{p}_j}\pi_j(\mathbf{p}_j,\mathbf{p}_{-j};\mu,\Sigma)$ into a vector valued function of L entries we get a vector field over $\mathbb{R}^L$ that we denote

$$[D_{\mathbf{p}_j}\pi_j(\mathbf{p}_j,\mathbf{p}_{-j};\mu,\Sigma)]_j = \mathbf{s}(\mathbf{p};\mu,\Sigma) + [\Lambda(\mathbf{p};\mu,\Sigma) - H \odot \Gamma(\mathbf{p};\mu,\Sigma)](\mathbf{p} - \boldsymbol{c}) \tag{7}$$

where H is an $L \times L$ *ownership matrix* with entries $H_{km} = \mathbf{1}[j(k) = j(m), (k,m) \in \mathcal{L}^2]$ and $\odot$ is the Hadamard product (Conlon and Gortmaker, 2020). Using the second partial derivatives of profits with respect to prices we get the Jacobian of (7) with respect to $\mathbf{p}$:

$$D_{\mathbf{p}}\left([D_{\mathbf{p}_j}\pi_j(\mathbf{p}_j,\mathbf{p}_{-j};\mu,\Sigma)]_j\right) \tag{8}$$

The arrays of second partial derivatives are more extensive and so relegated to Appendix A. Lemma 2.5 establishes the twice continuous differentiability of profit functions (4).

Lemma 2.5. *Fix μ and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. For all $j \in \mathcal{J}$:*

1. *Under mixed logit demand, j's profits are twice continuously differentiable for all $\mathbf{p} \in \mathbb{R}^L_+$ for all $\Sigma \in S_+$.*
2. *Under logit demand, j's profits are twice continuously differentiable for all $\mathbf{p} \in \mathbb{R}^L_+$.*

Proof. This and other proofs are relegated to Appendix A. For this proof $\underline{\alpha} \geq 0$ is sufficient, meaning Assumption 2.2 can be relaxed.

3 Convergence of Mixed Logit to Logit

In this section I prove that firm's profits with mixed logit demand, as well as their arrays of first and second partial derivatives with respect to prices, converge uniformly to their profits and arrays of partial derivatives under logit demand as the variance of random coefficients converges to zero.

The proof has three steps. First, I prove that, for every Σ, the set of simultaneously stationary points of a model is nonempty and in the interior of a box in R^L_+. Second, I prove the continuity of profits and their derivatives with respect to $(\mathbf{p},\Sigma)$. Finally, I prove the uniform convergence of profits and derivatives under mixed logit demand to their logit counterparts as Σ goes to $\mathbf{0}_\Sigma$.

3.1 Existence and Finiteness of Simultaneously Stationary Points

Lemma 3.1. *Fix μ and let Assumptions 2.1, 2.2, and 2.3 hold. Then, for each $\Sigma \in S \backslash \{\mathbf{0}_\Sigma\}$ there exists a vector $\overline{\mathbf{p}}(\Sigma) \in \mathbb{R}^L$, with $\overline{p}_l(\Sigma) > c_l$ for all $l \in \mathcal{L}$, such that the set of simultaneously stationary* *points of* (7) *is nonempty and contained in the interior of* $[\boldsymbol{c}, \overline{\mathbf{p}}(\Sigma)]$*. If Assumption 2.4 holds, there exists* $\overline{\mathbf{p}}(\mathbf{0}_\Sigma) \in \mathbb{R}^L$*, with* $\overline{p}_l(\mathbf{0}_\Sigma) > c_l$ *for all* $l \in \mathcal{L}$*, such that the set of simultaneously stationary points is nonempty and contained in the interior of* $[\boldsymbol{c}, \overline{\mathbf{p}}(\mathbf{0}_\Sigma)]$*.*

3.2 Continuity in $(\mathbf{p}, \Sigma)$

Lemma 3.2. *Fix* μ *and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then firm's profit functions and their arrays of first and second partial derivatives with respect to prices are continuous in* $(\mathbf{p}, \Sigma) \in \mathbb{R}^L \times S$*.*

Now we prove that we can constrain our attention to a single box in R_+^L. This requires an upper bound $R > 0$ on the norm of variance-covariance matrices. This bound is auxiliary to the proof, but poses no constraints on the set of models we consider, as it can be made arbitrarily large. It can therefore match any estimated variance-covariance matrix.[2] Define

$$S_R = \{\Sigma \in S : \|\Sigma\|_\infty \leq R\}.$$

Lemma 3.3. *Fix* μ *and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for every* $R > 0$ *there exists* $\overline{\mathbf{p}}(R) \in \mathbb{R}^L$ *such that, for all* $\Sigma \in S_R$*,* $\infty > \overline{p}_l(R) \geq \overline{p}_l(\Sigma) > c_l$ *for all* $l \in \mathcal{L}$*.*

Lemma 3.4 presents a uniform continuity result.

Lemma 3.4. *Fix* μ *and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all* $R > 0$*, for each* $j \in \mathcal{J}$*, firm profits, and their arrays of first and second partial derivatives with respect to* $\mathbf{p}$*, are uniformly continuous in* $(\mathbf{p}, \Sigma) \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$*.*

3.3 Uniform Convergence to Logit

The continuity result of Lemmas 3.2 and 3.4 mean that firms' profits under mixed logit demand, and their arrays of first and second partial derivatives, converge pointwise (for every $\mathbf{p}$ in a box $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$) to the profits and arrays of first and second partial derivatives under logit demand as $\Sigma \to \mathbf{0}_\Sigma$. Our existence and uniqueness result requires a stronger, uniform convergence result.

Lemma 3.5. *Fix* μ *and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all* $R > 0$*, restricting the domain of prices to* $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$*:*

1. *For all* $j \in \mathcal{J}$*, profit function* $\pi_j(\cdot; \mu, \Sigma)$ *converges uniformly to* $\pi_j(\cdot; \mu, \mathbf{0}_\Sigma)$ *as* $\Sigma \to \mathbf{0}_\Sigma$*.*
2. *The array of firms' profit functions' first partial derivatives,* $[D_{\mathbf{p}}\pi_j(\cdot; \mu, \Sigma)]_j$*, converges uniformly to* $[D_{\mathbf{p}}\pi_j(\cdot; \mu, \mathbf{0}_\Sigma)]_j$ *as* $\Sigma \to \mathbf{0}_\Sigma$*.*
3. *The array of firms' profit functions' second partial derivatives,* $[D^2_{\mathbf{pp}}\pi_j(\cdot; \mu, \Sigma)]_j$*, converges uniformly to* $[D^2_{\mathbf{pp}}\pi_j(\cdot; \mu, \mathbf{0}_\Sigma)]_j$ *as* $\Sigma \to \mathbf{0}_\Sigma$*.*

Appendix B provides a more general proof of uniform convergence that uses the joint continuity of functions in their variables and parameters. That result may be applied to this and other economic models.

[2] As pointed out by Berry et al. (1995), global uniqueness of equilibrium is not necessary for the estimation of parameters under mixed logit.

4 Existence of a Unique Bertrand-Nash Equilibrium

In this section I prove there is a family of mixed models that satisfy Assumptions 2.1, 2.2, and 2.3 that have a unique Bertrand-Nash equilibrium. A family is defined by a set of variance-covariance matrices $\Sigma \in S_+$. μ is fixed.

To do this I prove three results. First, there is a set of $\Sigma \in S_+$ such that, for all mixed logit models in that collection, the first-order condition system (7) has a unique simultaneously stationary point. Second, there is a set of $\Sigma \in S_+$ such that, for all mixed logit models in that collection, if a price vector $\mathbf{p}^*$ is an SSP, then all firms globally maximize profits at $\mathbf{p}^*_j$, given $\mathbf{p}^*_{-j}$. Finally, I prove the intersection of these sets contains a nonempty open ball in S_R.

4.1 Uniqueness of Simultaneously Stationary Points

Konovalov and Sándor (2010) use Kellogg (1976)'s proof of uniqueness of the Brouwer fixed point theorem to prove that a logit model with linear utility (Assumption 2.4) has a unique Bertrand-Nash equilibrium. This requires proving that (1) profit functions be twice-continuously differentiable, (2) finding a compact, convex set that contains all simultaneously stationary points of the first-order condition system in its interior, and (3) proving that the Jacobian determinant of that system is always positive in that convex compact set. Lemmas 2.5 and 3.3 prove the first two conditions.

It remains to prove that there are mixed logit models that satisfy Assumptions 2.1, 2.2, and 2.3 for which the Jacobian determinant of (7) is always positive in $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. Since the determinant of a matrix is a continuous function of the matrix entries, and since profits are smooth under our assumptions, we will use the proof of convergence of mixed logit to logit to prove this result.

Lemma 4.1. *Fix μ and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all $R > 0$, there exists an open ball in S_R, of radius $\varepsilon_1 > 0$ and centered in $\mathbf{0}_\Sigma$, $B_{S_R}(\mathbf{0}_\Sigma, \varepsilon_1)$, such that, for all $\Sigma \in B_{S_R}(\mathbf{0}_\Sigma, \varepsilon_1)\backslash\{\mathbf{0}_\Sigma\}$, the first-order condition system*

$$\mathbf{s}(\mathbf{p}; \mu, \Sigma) + [\Lambda(\mathbf{p}; \mu, \Sigma) - H \odot \Gamma(\mathbf{p}; \mu, \Sigma)](\mathbf{p} - \boldsymbol{c}) = 0$$

has a unique simultaneously stationary point.

4.2 Global Maximization at Simultaneously Stationary Points

The following result proves that at the unique simultaneously stationary point prices are such that all firms are maximizing their profits given all other prices.

Lemma 4.2. *Fix μ and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all $R > 0$, there exists a nonempty open ball in S_R, centered in $\mathbf{0}_\Sigma$ and of radius $\varepsilon_2 \in (0, \varepsilon_1]$, $B_{S_R}(\mathbf{0}_\Sigma, \varepsilon_2)$, such that for all $\Sigma \in B_{S_R}(\mathbf{0}_\Sigma, \varepsilon_2)\backslash\{\mathbf{0}_\Sigma\}$, letting $\mathbf{p}^*(\Sigma)$ be the unique simultaneously stationary point of (7) under that Σ,*

$$\mathbf{p}^*_j(\Sigma) = \arg \max_{\mathbf{p}'_j \in [\boldsymbol{c}_j, \overline{\mathbf{p}}_j(R)]} \pi_j(\mathbf{p}'_j, \mathbf{p}^*_{-j}(\Sigma); \mu, \Sigma) \qquad \forall j \in \mathcal{J}.$$

4.3 Existence and Global Uniqueness

Theorem 4.3. *Fix μ and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all $R > 0$, there exists a subset of S_R, $\mathcal{BN}$, with $\mathbf{0}_\Sigma \in \mathcal{BN}$ and $\mathcal{BN}$ containing a nonempty open ball in S_R, such that, for all $\Sigma \in \mathcal{BN} \backslash \{\mathbf{0}_\Sigma\}$, the corresponding mixed logit model has a unique Bertrand-Nash equilibrium.*

Theorem 4.3 imposes no constraints on the market share of the outside good, other than it must not be zero (Assumption 2.1). This matters because an existing conjecture is that if the share of the outside good is large enough, a unique Bertrand-Nash equilibrium exists.[3] Theorem 4.3 shows that any positive share for the outside good is consistent with some models having a unique equilibrium. An interesting result would be proving that the set of variance-covariance matrices that yield a unique equilibrium grows with the share of the outside good.

5 Relevance of the Variance Constraints

It would help to have lower bounds on the norms of the variance-covariance matrices such that a unique equilibrium exists. Here I point out a direction for finding such bounds. The first step is finding a lower bound on the norms of variance-covariance matrices such that (7) has a unique SSP. Since the proof of uniqueness uses the positive sign of the Jacobian determinant of (7) under logit demand, two quantities are useful for bounding the rate at which the determinant gets closer to zero: the smallest value of the Jacobian determinant over $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ under logit demand, and the Lipschitz constant of that determinant.[4]

Because $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ is compact, by the extreme value theorem and Lemma 5 of Konovalov and Sándor (2010) its minimum over $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ given $\Sigma = \mathbf{0}_\Sigma$ exists and is positive:

$$\min_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} \det\left(D_{\mathbf{p}}([D_{\mathbf{p}}\pi_j(\cdot; \mu, \mathbf{0}_\Sigma)]_j)\right) = \underline{d} > 0.$$

$\underline{d}$ is the smallest distance the determinant must travel before becoming non-positive. Thus we need to bound the rate at which it changes with $(\mathbf{p}, \Sigma)$. If a finite Lipschitz constant C_L exists, it can fulfill that role. With $\underline{d}$ and C_L in hand a lower bound on the norms of the variance-convariance matrices that yield a unique SSP may be computed. Here I provide steps to prove a finite C_L exists.

If the determinant is continuously differentiable in in interior of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$ and has a continuously differentiable extension in $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$, then it is Lipschitz continuous in $(\mathbf{p}, \Sigma) \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$. The determinant of $D_{\mathbf{p}}([D_{\mathbf{p}}\pi_j(\cdot; \mu, \mathbf{0}_\Sigma)]_j)$ is a function of the second derivatives of firms' profits with respect to prices, so continuous differentiability of the determinant requires at least three times continuous

[3] See Aksoy-Pierson et al. (2013) and sources cited therein.

[4] An interesting point, not dealt with here, is that the sets S_R and open balls in it depend on the choice of matrix norm. All norms in finite-dimensional spaces are equivalent, but they generate different open balls. The choice of matrix norm does not affect the set of variance-covariance matrices that yield a unique equilibrium, but may affect what we can say about bounds of the set of such matrices. My choice of the maximum norm comes from the search for consistency with (Morrow and Skerlos, 2010a,b), who use the maximum norm.

differentiability of profits with respect to prices.[5] [6]

Lemma 5.1. *Fix μ and let Assumptions 2.1, 2.2, 2.3, and 2.4 hold. Then, for all $R > 0$, for all $j \in \mathcal{J}$, $\pi_j(\mathbf{p}; \mu, \Sigma)$ is three times continuously differentiable with respect to $\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ for all $\Sigma \in S_R$.*

6 Relaxing Assumptions

In this section I go through Assumptions 2.1 through 2.3, and show that they either match assumptions in classical mixed logit models (e.g., Berry (1994) and McFadden and Train (2000)), or can be relaxed/generalized. McFadden and Train (2000) is germane because its approximation theorem shows that under some regularity conditions any random utility maximization model can be approximated arbitrarily well with a sufficiently rich mixed logit model.

Some arguments in this section refer to steps in the proofs in Appendix A, which the reader may safely skip.

6.1 Assumption 2.1

Linearity of utility in $(\alpha_i, \boldsymbol{\beta}_i, p_l, \mathbf{x}_l)$. McFadden and Train (2000)'s approximation theorem uses dot products $(\alpha_i, \boldsymbol{\beta}_i)^T(p_l, \mathbf{x}_l)$ as the argument in the logit choice probabilities whose mixing is used to approximate any RUM model arbitrarily well.

Linearity of utility price. The previous argument ignores the possibility of interactions between p and other characteristics in $\mathbf{x}$. Linearity in price, however, has the advantage that if the distribution of income is approximately log-normal, linear-in-price utility is a linear first-order approximation of models with income constraints, like that in Berry et al. (1995) (Berry et al., 1999).

6.2 Assumption 2.2

$\underline{\alpha} > 0$ is used in the proof of Lemma 3.1 to ensure that

$$\sup\left\{||\tilde{\Omega}(\mathbf{p})(\mathbf{p} - \boldsymbol{c})||_\infty : \mathbf{p} \in (0, \varsigma_*)^J\right\} < \infty.$$

Since the pdf of the shifted log-normal distribution is continuous in parameters, if $\underline{\alpha} > 0$ is arbitrarily small, the inequality will hold with $\underline{\alpha} = 0$ for some subset of $(\mu_\alpha, \sigma_\alpha) \in \mathbb{R} \times \mathbb{R}_{++}$. This set, however, cannot be characterized purely from primitives.

[5] The question of how to find $\underline{d}$ is not addressed in this version of the paper. Gradient descent, however, is a good candidate algorithm, because (1) it uses the differentiability result of Lemma 5.1 and the convexity of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$, (2) it is fast, and (3) it reliably converges to (at least local) minimizers if the objective function is twice continuously differentiable. Twice continuous differentiability of the determinant with respect to prices requires four times continuous differentiability of firms' profit functions with respect to $\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. See Appendix A for how to get that result. Given linear utility, it is a straightforward, if tedious, extension of the proof of Lemma 5.1.

[6] Note that Lemma 3.3 gives explicit expressions for $\overline{\mathbf{p}}(R)$ (given R), so the search spaces for $\underline{d}$ and C_L are known. We may be able to further reduce it for the search for $\underline{d}$, depending on $\overline{\mathbf{p}}(\mathbf{0}_\Sigma)$, whose construction is also shown in the proof of Lemma 3.3. Since we have the estimated variances, we know the minimum R we need. These considerations reduce the computational cost, as the researcher knows the search space exactly.

An alternative point is that $\underline{\alpha}$ can be made arbitrarily close to zero. Thus, because of the continuity of the shifted log-normal's pdf, the shifted log-normal distribution of α_i converges to a log-normal as $\underline{\alpha} \to 0$.

6.3 Assumption 2.3

Log-normal distribution of $(\alpha_i - \underline{\alpha})$**.** As mentioned above, a linear-in-price utility model with log-normally distributed price sensitivity approximates a model with nonlinear utility and income constraints if the distribution of income is approximately log-normal. With $\underline{\alpha}$ arbitrarily close to zero a model that satisfies Assumption 2.3 also achieves this approximation.

Jointly normal distribution of $\boldsymbol{\beta}_i$**.** Using a normal distribution allows us to index a mixed logit model using the variance-covariance matrix, because given μ the normal and log-normal distributions are determined by this matrix. Moreover, a normal distribution greatly simplifies the proof of Lemma 3.2, because it allows us to transform all distributions in the model to a standard normal via a simple change of variables. None of these steps are essential to any of the proofs, however. For example, one could use mixtures of normals and log-normals, or latent classes.[7] The cost of using alternative distributions is having to use other (and perhaps more) deep parameters for the mixing distribution, and an increase in the complexity of the proofs.

Independence of $(\alpha_i - \underline{\alpha})$ **and** $\boldsymbol{\beta}_i$**.** Independence between price sensitivities and other dimensions of tastes is a strong assumption. Fortunately, it is only used to prove the existence of some (finite) upper and (positive) lower bounds throughout Appendix A with closed form. These bounds would increase in complexity, but would still be positive and finite, without the independence assumption, so Assumption 2.3.3 is not essential. Nevertheless, Assumption 2.3.3 has been used in empirical work: see Berry et al. (1995, 1999).

7 Conclusion

In this paper I prove the uniform convergence of linear utility mixed logit models to logit as the variances of random coefficients go to zero. Then I use the existence of a unique Bertrand-Nash equilibrium under logit demand to prove that, at least for some distributions of random coefficients, linear utility models with mixed logit demand have a unique Bertrand-Nash equilibrium. The key step in the proof is a non-trivial uniform convergence result.

Moving forward, a first step is finding explicit lower bounds on the norms of variance-covariance matrices such that the equilibrium is unique. The objective is to extend this initial proof to larger sets of mixed logit models. Extension of resukts to at least local uniqueness of equilibria is also desirable.

[7] The use of normal and log-normal distributions has the added benefit that it matches the model specifications relevant for empirical work.

Appendices

A Main Proofs

A.1 Morrow and Skerlos' Assumptions

Since the proofs depend on results under assumptions stated in Morrow and Skerlos (2010a,b), I state those assumptions first. I follow the notation of those papers when possible.

A.1.1 Preliminaries for Mixed Logit

Morrow and Skerlos (2010a) show that a solution of system of all firms' FOCs is a fixed point of the ζ markup equation:

$$\mathbf{p} = \mathbf{c} + \zeta(\mathbf{p})$$

with

$$\zeta(\mathbf{p}) = \Lambda(\mathbf{p})^{-1}[H \odot \Gamma(\mathbf{p})](\mathbf{p} - \boldsymbol{c}) - \Lambda(\mathbf{p})^{-1}\mathbf{s}(\mathbf{p}). \tag{9}$$

The expression

$$F_\zeta(\mathbf{p}) = \mathbf{p} - \mathbf{c} - \zeta(\mathbf{p}) \tag{10}$$

defines a vector field over P whose zeros are simultaneously stationary points (SSPs): points where the FOCs of all firms are satisfied. Finally define

$$\Omega_j(\mathbf{p}) = \Lambda_j(\mathbf{p})^{-1}\Gamma_j(\mathbf{p})^T \in \mathbb{R}^{L_j \times L_j} \quad \forall j \tag{11}$$

$$\widetilde{\Omega}(\mathbf{p}) = \Lambda(\mathbf{p})^{-1}(H \odot \Gamma(\mathbf{p}))^T \in \mathbb{R}^{L \times L}. \tag{12}$$

A.1.2 Assumptions for Mixed Logit

Assumption A.1. *For all $j \in 0 \cup \mathcal{L}$ there exist functions $w_l : P \times \Theta \to [-\infty, \infty)$ and $v : \Theta \to \mathbb{R}$ such that $u_l(p_l, \theta) = w_l(p, \theta) + v_l(\theta)$, with $w_0 = 0$.*

Assumption A.2. *There exists $\varsigma : \Theta \to (0, \infty]$ such that, for all $l \in \mathcal{L} \cup \{0\}$, $w_l : [0, \infty] \times \Theta \to [-\infty, \infty)$ satisfies ρ-almost everywhere (a.e.) $\boldsymbol{\theta} \in \Theta$:*

1. *$w_l(\cdot, \boldsymbol{\theta}) : (0, \varsigma(\boldsymbol{\theta})) \to [-\infty, \infty)$ is concave, twice continuously differentiable, strictly decreasing, and finite.*
2. *$w_l(p_l, \boldsymbol{\theta}) = -\infty$ if $p_l \geq \varsigma(\boldsymbol{\theta})$.*
3. *$w_l(p_l, \boldsymbol{\theta}) \downarrow -\infty$ as $p_l \uparrow \varsigma(\boldsymbol{\theta})$.*
4. *$v_l(\boldsymbol{\theta})$ is finite.*

Function ς is a type-dependent choke price, which may be infinite. The linear utility model implicitly assumes $\varsigma(\boldsymbol{\theta}) = \infty\ \forall \theta \in \Theta$, but it is a first-order approximation of finite income models (Berry et al., 1999; Conlon and Gortmaker, 2020). Let $\varsigma_* = ess\sup \varsigma(\boldsymbol{\theta})$, the *essential supremum* of ς. Therefore, $\{\boldsymbol{\theta} : \varsigma(\boldsymbol{\theta}) \geq \varsigma_*\}$ has ρ-measure zero.

Assumption A.3. *For all l there exists some $r_l : \Theta \to (1, \infty)$ and $\bar{p}_l : \Theta \to P$ satisfying*

$$\sup \{p\rho(\{\boldsymbol{\theta} : \bar{p} > p\}) : p \in (0, \varsigma_*)\} < \infty \tag{13}$$

such that

$$u_l(p_l, \boldsymbol{\theta}) \leq r_l(\boldsymbol{\theta}) \ln p_l + v_0(\boldsymbol{\theta}) \tag{14}$$

for all $p_l \geq \bar{p}_l(\boldsymbol{\theta})$, ρ-a.e.

Assumption A.4. *Let $k \in \mathcal{L}$ be arbitrary and define $\psi_k : \Theta \times P \to P$ by*

$$\psi_k(p_k, \boldsymbol{\theta}) = \begin{cases} |D_{p_k} w_k|(p_k, \boldsymbol{\theta}) \left(\frac{e^{u_k(p_k,\boldsymbol{\theta})}}{e^{u_0(\boldsymbol{\theta})} + e^{u_k(p_k,\boldsymbol{\theta})}}\right) & \text{if } p_k < \varsigma(\boldsymbol{\theta}) \\ 0 & \text{if } p_k \geq \varsigma(\boldsymbol{\theta}). \end{cases} \tag{15}$$

Let

1. *$\psi_k(\cdot, \boldsymbol{\theta}) : (0, \varsigma_*) \to P$ is continuous for μ-a.e. That is, $\psi_k(q, \boldsymbol{\theta}) \to \psi_k(p, \boldsymbol{\theta})$ as $q \to p$ for all $p \in P$.*
2. *$\psi_k(p, \cdot) : \Theta \to P$ is uniformly $\mu-$integrable for all q in some neighborhood of $p \in (0, \varsigma_*)$. That is, there exists some $\varphi : \Theta \to [0, \infty)$, with $\int \varphi(\boldsymbol{\theta}) d\rho(\boldsymbol{\theta}) < \infty$ (which may depend on k and p), such that $\psi_k(q, \boldsymbol{\theta}) \leq \varphi(\boldsymbol{\theta})$ for all q in some neighborhood of p.*

Assumption A.5.

$$\sup \left\{ ||\tilde{\Omega}(\mathbf{p})(\mathbf{p} - \boldsymbol{c})||_\infty : \mathbf{p} \in (0, \varsigma_*)^J \right\} < \infty. \tag{16}$$

Assumption A.6. *Let $\varsigma_* = \infty$ and*

$$\lim_{M \uparrow \infty} \sup \left\{ \frac{||\Lambda(\mathbf{p})^{-1} \mathbf{s}(\mathbf{p})||_\infty}{||\mathbf{p}||_\infty} : \mathbf{p} \in P^J, \, ||\mathbf{p}||_\infty \geq M \right\} = \delta \in [0, 1). \tag{17}$$

The limit is a non-increasing sequence of negative numbers, so it exists (Morrow and Skerlos, 2010a).

Assumption A.7. *Let*

1. *For all $l \in \mathcal{L}$, $w_l(\cdot, \boldsymbol{\theta}) : (0, \varsigma_*) \to \mathbb{R}$ is twice continuously differentiable $\rho-$a.e. $\boldsymbol{\theta} \in \Theta$.*
2. *For all $l \in \mathcal{L}$, for all $p \in (0, \varsigma_*)$, $|D^2_{qq} w_l(q, \cdot) + (D_q w_l(q, \cdot))^2| \exp\{u_l(q, \cdot) - u_0(\cdot)\} : \Theta \to [0, \infty)$ is uniformly $\rho-$integrable for all q in some neighborhood of p.*
3. *For all $l \in \mathcal{L}$, for all $p, p' \in (0, \varsigma_*)$*

$$|D_q w_l(q, \cdot)| \exp\{u_l(q, \cdot) - u_0(\cdot)\} \exp\{u_l(q', \cdot) - u_0(\cdot)\} |D_{q'} w_l(q', \cdot)| : \Theta \to [0, \infty)$$

is uniformly $\rho-$integrable for all (q, q') in some neighborhood of (p, p').

A.1.3 Assumptions for Logit

Since product characteristics play no role in our proofs, here we suppress them from our notation, and adjust the notation from Morrow and Skerlos (2010b) accordingly.

Assumption A.8. *There is a function* $w : [0, \infty) \to (-\infty, \infty)$*, and constant* $v_l \in (-\infty, \infty)$ *for all* $l \in \mathcal{L} \cup \{0\}$*, such that* $u_l(p_l) = w(p) + v_l$ *for all* $l \in \mathcal{L} \cup \{0\}$*. Moreover* w *is*

1. *strictly decreasing,*
2. *twice continuously differentiable,*
3. $\lim_{p\uparrow\infty} w(p) = -\infty$*, and*
4. $w(\infty) = -\infty$.

Assumption A.9. w *eventually decreases sufficiently quickly: there exists some* $r > 1$*, and some* $\overline{p} \in [0, \infty)$ *such that* $D_p w(p) \leq -r/p$ *for all* $p > \overline{p}$

Assumption A.10. w *has sub-quadratic second derivatives:* $D^2_{pp} w(\cdot)/(D_p w(\cdot))^2 < 1$*.*

Assumption A.11. w *is eventually log-bounded: There exist* $r > 1$*,* κ*, and some* $\overline{p} \in [0, \infty)$ *such that* $w(p) \leq -r \ln(p) + \kappa$ *for all* $p > \overline{p}$*.*

A.2 Proof of Lemma 2.5

Strategy of proof. Proposition 2.4 of Morrow and Skerlos (2010a) shows that if Assumptions A.1, A.2, and A.4 hold then market shares are continuously differentiable. If Assumption A.7 also holds, by Proposition 3.10 of that paper they are twice continuously differentiable. Continuous differentiability of market shares, they show, implies continuous differentiability of profit functions, obtaining the result. So, proving part 1 of Lemma 2.5 reduces to proving Assumptions A.1, A.2, and A.4 hold. Lemma 3.3 of Morrow and Skerlos (2010b) shows that if Assumption A.8 holds, then twice continuous differentiability holds for the logit model. To simplify notation, we drop μ and Σ are arguments for the functions.

To explain the proof, I first present the arrays of second partial derivatives, if they exist. By Proposition 3.9 of Morrow and Skerlos (2010a), firm's profit functions' Hessians can be constructed using the following expressions:

$$\phi_{k,l}(\mathbf{p}) = \int \alpha_i^2 s_k^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) d\rho(\widetilde{\boldsymbol{\theta}}_i) \tag{18}$$

$$\psi_{k,l}(\mathbf{p}) = \int \alpha_i^2 s_k^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) \left[\sum_{h \in L_{j(k)}} (p_h - c_h) s_h^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) \right] d\rho(\widetilde{\boldsymbol{\theta}}_i) \tag{19}$$

$$\chi_k(\mathbf{p}) = \frac{1}{2} \int \alpha_i^2 s_k^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) \left[(p_k - c_k) + \sum_{h \in L_{j(k)}} (p_h - c_h) s_h^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i) \right] d\rho(\widetilde{\boldsymbol{\theta}}_i). \tag{20}$$

Now let

$$\eta_{k,l}(\mathbf{p}) = \mathbf{1}[k = l][\lambda_k(\mathbf{p}) + \chi_k(\mathbf{p})] - \gamma_{k,l}(\mathbf{p}) - (p_k - c_k)\phi_{k,l}(\mathbf{p}). \tag{21}$$

where $\lambda_k(\mathbf{p})$ comes from eq. (5) and $\gamma_{k,l}(\mathbf{p})$ comes from eq. (6). Then, by Proposition 3.9 of Morrow and Skerlos (2010a), the second partial derivative has elements

$$\frac{\partial^2 \hat{\pi}_j(\mathbf{p})}{\partial p_k \partial p_l} = \eta_{k,l}(\mathbf{p}) + 2\psi_{k,l}(\mathbf{p}) + \mathbf{1}[j(k) = j(l)]\eta_{l,k}(\mathbf{p}). \tag{22}$$

Proof of part 1. Assumptions 2.1, A.2, and 2.3 trivially imply A.1 and A.2 under $\varsigma_* = \infty$. Under A.2.1, A.4.1 only requires that $\psi_k(\boldsymbol{\theta}, \mathbf{p}_j) \to 0$ as $p \uparrow \varsigma(\boldsymbol{\theta})$ for ρ-a.e. $\boldsymbol{\theta}$ (Morrow and Skerlos, 2010a). This follows from Assumptions 2.1.

Finally, for the uniform integrability condition, note that $e^{u_k(p_k,\boldsymbol{\theta}_i)}/(e^{u_0(\boldsymbol{\theta}_i)} + e^{u_k(p_k,\boldsymbol{\theta}_i)}) \in (0, 1)$ for all $k \in \mathcal{L}$, so $\alpha_i e^{u_k(p_k,\boldsymbol{\theta}_i)}/(e^{u_0(\boldsymbol{\theta}_i)} + e^{u_k(p_k,\boldsymbol{\theta}_i)}) < \alpha_i$ for all k, for all α_i. Since the first moment of α_i is finite (Assumption 2.3.1), we have our integrable dominating function. This completes the proof of continuous differentiability.

Moving on to twice continuous differentiability, part 1 of Assumption A.7 is satisfied by (1) (Assum. 2.1). For part 2, $(D_q w_l(q, \boldsymbol{\theta}_i))^2 = \alpha_i^2$ and $D_{qq}^2 w_l(q, \boldsymbol{\theta}_i) = 0$ for all l, for all $\boldsymbol{\theta}_i$ by (1). So, we care about $g_p(\boldsymbol{\theta}_i) = \alpha_i^2 \exp(\mathbf{x}_l^T \boldsymbol{\beta}_i + \xi_l - \alpha_i p)$. Fix $p > 0$, choose $\epsilon \in (0, p)$, and let $m := p - \epsilon$. We will consider q in a ball in $\mathbb{R}$ centered at p with radius ϵ. Then, $q \geq m$, so

$$g_q(\boldsymbol{\theta}_i) \leq \alpha_i^2 \exp(\xi_l) \exp(\mathbf{x}_l^T \boldsymbol{\beta}_i) \exp(-\alpha_i m).$$

Note that $\alpha_i^2 \exp(-\alpha_i m)$, $\alpha_i \geq 0$ is bounded:

$$\sup_{\alpha_i \geq 0} \alpha_i^2 \exp(-\alpha_i m) = \frac{4}{e^2 m^2} \tag{23}$$

so

$$g_q(\boldsymbol{\theta}_i) \leq \frac{4 e^{\xi_l} e^{\mathbf{x}_l^T \boldsymbol{\beta}_i}}{e^2 m^2} =: G(\boldsymbol{\theta}_i)$$

for all $q \in (p - \epsilon, p + \epsilon)$. Because $\boldsymbol{\beta}_i$ is multivariate normal, from the properties of the multivariate log-normal distribution we get

$$\mathbb{E}[\exp(\mathbf{x}_l^T \boldsymbol{\beta}_i)] = \exp\left(\mathbf{x}_l^T \overline{\boldsymbol{\beta}} + \frac{1}{2}\mathbf{x}_l^T \Sigma_{\boldsymbol{\beta}} \mathbf{x}_l\right) < \infty$$

so $\mathbb{E}[G(\boldsymbol{\theta}_i)] < \infty$. So, the function q for part 2 of Assumption A.7 is uniformly μ-integrable for all p for all q in a neighborhood of p. For part 2 of Assum A.7, pick p, p', and $\epsilon \in (0, p)$, $\epsilon' \in (0, p')$ and use the balls $(p - \epsilon, p + \epsilon)$ and $(p' - \epsilon', p' + \epsilon')$. Let $q \in (p - \epsilon, p + \epsilon)$ and $q' \in (p' - \epsilon', p' + \epsilon')$. Let $m := (p - \epsilon) + (p' - \epsilon') > 0$. Then $q + q' \geq m$ The relevant function for our proof is

$$\begin{aligned} h_{q,q'}(\boldsymbol{\theta}_i) &= \alpha_i^2 \exp(\mathbf{x}_l^T \boldsymbol{\beta}_i + \xi_l - \alpha_i q) \exp(\mathbf{x}_l^T \boldsymbol{\beta}_i + \xi_l - \alpha_i q') \\ &= \alpha_i^2 \exp(2\mathbf{x}_l^T \boldsymbol{\beta}_i + 2\xi_l - \alpha_i (q + q')) \end{aligned}$$

which has the following bound:

$$h_{q,q'}(\boldsymbol{\theta}_i) \leq \exp(2\xi_l)\exp(2\mathbf{x}_l^T\boldsymbol{\beta}_i)\alpha_i^2\exp(-m\alpha_i) =: H(\boldsymbol{\theta}_i)$$

By (23) we obtain

$$h_{q,q'}(\boldsymbol{\theta}_i) = \frac{4\exp(2\xi_l)}{e^2m^2}\exp(2\mathbf{x}_l^T\boldsymbol{\beta}_i)$$

By the normality assumption in Assum 2.3 we get

$$\mathbb{E}[\exp(2\mathbf{x}_l^T\boldsymbol{\beta}_i)] = \exp\left(2\mathbf{x}_l^T\overline{\boldsymbol{\beta}} + \frac{1}{2}4\mathbf{x}_l^T\Sigma_{\boldsymbol{\beta}}\mathbf{x}_l\right) < \infty$$

Therefore H is integrable, and the family $\{h_{q,q'} : q, q' \text{ in neighborhoods of } p, p'\}$ is dominated by a common integrable random variable, yielding uniform μ-integrability.

Proof of part 2. Follows trivially from Assumption A.8. ■

A.3 Proof of Lemma 3.1

We need to prove that Assumptions A.1 through A.11 hold under Assumptions 2.1 through 2.4. We skip assumptions already proven in the proof of Lemma 2.5. Note that the proof in Morrow and Skerlos (2010a) is about the set of SSPs of F_ζ, not of the vector field of FOCs. However, F_ζ is just a transformation of the FOCs, and they have the same sets of SSPs.

Part 1: Proof for mixed logit. Assumption A.3. If $\varsigma_* = \infty$, concavity of w_l in p_l for all $l \in \mathcal{L}$ (Assumption A.1) implies Assumption A.3 holds (Morrow and Skerlos, 2010a).

Assumption A.5. From (12) we get that

$$\widetilde{\omega}_{lk}(\mathbf{p}) = \frac{\int \alpha_i s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)\, s_k^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)}{\int \alpha_i s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)}$$

The boundedness of market shares from above by 1 yields the following upper bound on the numerator:

$$s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)s_k^{Lo}(\mathbf{p},\boldsymbol{\theta}_i) \leq \exp(\mathbf{x}_l\boldsymbol{\beta}_i + \xi_l - \alpha_i p_l)\exp(\mathbf{x}_k\boldsymbol{\beta}_i + \xi_k - \alpha_i p_k)$$
$$\Rightarrow \int \alpha_i s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)s_k^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i) \leq \int \alpha_i \exp(\mathbf{x}_l\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k\boldsymbol{\beta}_i + \xi_k)\exp(-\alpha_i(p_l+p_k))d\rho(\widetilde{\boldsymbol{\theta}}_i).$$

By the independence part of Assumption 2.3

$$\begin{aligned}
&\int \alpha_i \exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)\exp(-\alpha_i(p_l+p_k))d\rho(\widetilde{\boldsymbol{\theta}}_i) \\
&= \mathbb{E}_{\widetilde{\boldsymbol{\theta}}_i}\left[\alpha_i\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)\exp(-\alpha_i(p_l+p_k))\right] \\
&= \mathbb{E}_{\boldsymbol{\beta}_i}\left[\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)\right]\mathbb{E}_{\alpha_i}\left[\alpha_i\exp(-\alpha_i(p_l+p_k))\right].
\end{aligned}$$

The joint normality of $\boldsymbol{\beta}_i$ yields

$$\mathbb{E}[\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)] < \infty.$$

By Assumption 2.2

$$\begin{aligned}
&\exp(-\alpha_i(p_l + p_k) \leq \exp(-\underline{\alpha}(p_l + p_k)) \quad \forall \alpha_i \\
\Rightarrow\ & \mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\alpha_i(p_l + p_k))\right] \leq \mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\underline{\alpha}p_k)\exp(-\alpha_i p_l)\right] = \mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\alpha_i p_l)\right]\exp(-\underline{\alpha}p_k) \\
&\leq \mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\underline{\alpha}(p_l + p_k))\right] \\
&= \exp(-\underline{\alpha}(p_l + p_k))\,\mathbb{E}_{\alpha_i}\left[\alpha_i\right] < \infty
\end{aligned}$$

where the finiteness comes from the finiteness of the moments of the shifted log-normal distribution. Now we get a lower bound on the numerator: since $p_l \geq 0$ for all L

$$\exp(u_0) + \sum_k \exp(u_k(p_k, \boldsymbol{\theta}_i)) \leq \exp(u_0) + \sum_k \exp(\mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)$$

Therefore

$$s_l^{Lo}(\mathbf{p}) \geq \exp(-\alpha_i p_l)\frac{\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l)}{\exp(u_0) + \sum_k \exp(\mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)}$$

Define

$$\begin{aligned}
Q_l(\underbrace{\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l}_{\equiv V_l(\boldsymbol{\theta}_i)}) &:= \frac{\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l)}{\exp(u_0) + \sum_k \exp(\mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)} \in (0,1) \\
&\Rightarrow \mathbb{E}_{\boldsymbol{\beta}_i}[Q_l(V_l(\boldsymbol{\theta}_i))] \in (0,1).
\end{aligned}$$

Therefore

$$\begin{aligned}
\int \alpha_i s_l^{Lo}(\mathbf{p}, \boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i) &= \mathbb{E}_{\widetilde{\boldsymbol{\theta}}_i}\left[\alpha_i \frac{\exp(-\alpha_i p_l + \mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l)}{\exp(u_0) + \sum_k \exp(-\alpha_i p_k + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)}\right] \\
&\geq \mathbb{E}_{\widetilde{\boldsymbol{\theta}}_i}\left[\alpha_i \exp(-\alpha_i p_l) Q_l(V(\boldsymbol{\theta}_i))\right] \\
&\geq \underline{\alpha}\,\mathbb{E}_{\alpha_i}\left[\exp(-\alpha_i p_l)\right]\mathbb{E}_{\boldsymbol{\beta}_i}\left[Q_l(V(\boldsymbol{\theta}_i))\right] > 0.
\end{aligned}$$

Note that $\mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\alpha_i p_l)\right] > 0$ for all $p_l \geq 0$. From these upper and lower bounds we have

$$\begin{aligned}
\widetilde{\omega}_{lk}(\mathbf{p}) &\leq \frac{\mathbb{E}[\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)]\,\mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\alpha_i p_l)\right]\exp(-\underline{\alpha}p_k)}{\mathbb{E}_{\alpha_i}\left[\alpha_i \exp(-\alpha_i p_l)\right]\mathbb{E}_{\boldsymbol{\beta}_i}\left[Q_l(V(\boldsymbol{\theta}_i))\right]} \\
&\leq \underbrace{\frac{\mathbb{E}[\exp(\mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l + \mathbf{x}_k^T\boldsymbol{\beta}_i + \xi_k)]}{\mathbb{E}_{\boldsymbol{\beta}_i}\left[Q_l(V(\boldsymbol{\theta}_i))\right]}}_{\equiv C_{lk} > 0}\exp(-\underline{\alpha}p_k). \qquad (24)
\end{aligned}$$

Going entry-by-entry through $\widetilde{\Omega}(\mathbf{p})(\mathbf{p}-\boldsymbol{c})$ we get

$$\widetilde{\omega}_{lk}(\mathbf{p})(p_k - c_k) \leq C_{lk}\exp(-\underline{\alpha} p_k)(p_k - c_k) \leq C_{lk}\exp(-\underline{\alpha} p_k)p_k$$

Note that $\sup_{p\in\mathbb{R}_+}\exp(-\underline{\alpha} p_k)p_k = 1/(e\underline{\alpha})$, so

$$\widetilde{\omega}_{lk}(\mathbf{p})(p_k - c_k) \leq \frac{C_{lk}}{e\underline{\alpha}} < \infty \quad \forall l,k \quad \forall \mathbf{p}\in\mathbb{R}^L.$$

Assumption A.6. From (3) and (5) we get, for each l

$$\begin{aligned} |\Lambda_{ll}(\mathbf{p})^{-1}s_l(\mathbf{p})| &= \frac{\int s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)}{\int \alpha_i s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)} \\ &\leq \frac{\int s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)}{\int \underline{\alpha} s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)d\rho(\widetilde{\boldsymbol{\theta}}_i)} = \frac{1}{\underline{\alpha}} \quad \forall \mathbf{p}\in\mathbb{R}^L_+. \end{aligned} \tag{25}$$

Since $\|\mathbf{p}\|_\infty = \infty$ as any $p_l \to \infty$, we get that Assumption A.6 is satisfied with $\delta = 0$.

Part 2: Proof for logit. We must prove Assumptions A.9, A.10, and A.11 hold.

Assumption A.9. If $w(p)$ is concave and strictly decreasing, it satisfies A.A10 (Morrow and Skerlos, 2010b).

Assumption A.10. Follows from $D^2_{pp}w(\cdot) = 0$ for all l for all p_l, and $D_p w(p) = -\overline{\alpha}$ for all l for all p_l.

Assumption A.11. Follows from $D_p w(p) = \overline{\alpha}$ for all l for all p_l. ■

A.4 Proof of Lemma 3.2

We will represent the random coefficients as functions of a standard-normally distributed random vector. Let $\boldsymbol{Z}$ be a random vector of $\dim(\Theta)$ elements, indexed 1 through $\dim(\Theta)$. Define

$$\alpha(z_1,\Sigma) = \underline{\alpha} + \exp(\mu_\alpha + z_1\sigma_\alpha) \quad \text{and} \quad \boldsymbol{\beta}(\boldsymbol{Z}_{-1},\Sigma) = \overline{\boldsymbol{\beta}} + \Sigma^{\frac{1}{2}}_{\boldsymbol{\beta}}\boldsymbol{Z}_{-1} \tag{26}$$

where $\Sigma^{\frac{1}{2}}_{\boldsymbol{\beta}}$ is the principal matrix square root. Let $\boldsymbol{\theta}_i(\boldsymbol{Z},\Sigma) \equiv (\alpha_i(z_1,\Sigma), \boldsymbol{\beta}_i(\boldsymbol{Z}_{-1},\Sigma))$. Therefore we can rewrite market shares as

$$s_l(\mathbf{p};\mu,\Sigma) = \int \frac{\exp(-\alpha(z_1,\Sigma)p_l + \mathbf{x}_l^T\boldsymbol{\beta}(\boldsymbol{Z}_{-1},\Sigma) + \xi_l)}{\exp(u_0) + \sum_k \exp(-\alpha(z_1,\Sigma)p_k + \mathbf{x}_k^T\boldsymbol{\beta}(\boldsymbol{Z}_{-1},\Sigma) + \xi_k)}\phi(\boldsymbol{Z})d\boldsymbol{Z}$$

where ϕ is the pdf of a (multivariate) standard normal. This follows because by the chain rule and independence in Assumption 2.3. Under Assumption 2.3 we can write

$$d\rho(\widetilde{\boldsymbol{\theta}}_i;\mu,\Sigma) = \underbrace{\frac{1}{(\alpha_i-\underline{\alpha})\sigma_\alpha\sqrt{2\pi}}\exp\left[-\frac{(\ln(\alpha_i-\underline{\alpha})-\mu_\alpha)^2}{2\sigma_\alpha^2}\right]}_{f_\alpha()} d\alpha_i \\ \times \underbrace{(2\pi)^{\frac{-K}{2}}\det(\Sigma_{\boldsymbol{\beta}})^{\frac{-1}{2}}\exp\left[-\frac{1}{2}(\boldsymbol{\beta}_i-\overline{\boldsymbol{\beta}})^T\Sigma_{\boldsymbol{\beta}}^{-1}(\boldsymbol{\beta}_i-\overline{\boldsymbol{\beta}})\right]}_{f_{\boldsymbol{\beta}}()} d\boldsymbol{\beta}_i \tag{27}$$

From (26) we get

$$\frac{d\alpha}{dz} = \sigma_\alpha\exp(\mu_\alpha+\sigma z_i) = \sigma_\alpha(\alpha-\underline{\alpha}) \Rightarrow d\alpha = \sigma_\alpha(\alpha-\underline{\alpha})dz \\ \frac{d\boldsymbol{\beta}}{d\boldsymbol{Z}} = |\det(\Sigma_{\boldsymbol{\beta}}^{1/2})| \Rightarrow d\boldsymbol{\beta} = |\det(\Sigma_{\boldsymbol{\beta}}^{1/2})|d\boldsymbol{Z} \tag{28}$$

Combining (26), (27), and (28), and the fact that $\alpha(z,\Sigma)-\underline{\alpha} = \exp(\mu_\alpha+\sigma_\alpha z)$ we get

$$\begin{aligned}&\frac{1}{(\alpha-\underline{\alpha})\sigma_\alpha\sqrt{2\pi}}\exp\left[-\frac{(\ln(\alpha_i-\underline{\alpha})-\mu_\alpha)^2}{2\sigma_\alpha^2}\right]d\alpha\\ =&\frac{1}{(\alpha-\underline{\alpha})\sigma_\alpha\sqrt{2\pi}}\exp\left[-\frac{(\mu_\alpha+\sigma_\alpha z-\mu_\alpha)^2}{2\sigma_\alpha^2}\right]\sigma_\alpha(\alpha-\underline{\alpha})dz\\ =&\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{z^2}{2}\right]dz = \phi(z)dz\end{aligned}$$

and

$$\begin{aligned}&(2\pi)^{\frac{-K}{2}}\det(\Sigma_{\boldsymbol{\beta}})^{\frac{-1}{2}}\exp\left[-\frac{1}{2}(\boldsymbol{\beta}_i-\overline{\boldsymbol{\beta}})^T\Sigma_{\boldsymbol{\beta}}^{-1}(\boldsymbol{\beta}_i-\overline{\boldsymbol{\beta}})\right]d\boldsymbol{\beta}_i\\ =&(2\pi)^{\frac{-K}{2}}\det(\Sigma_{\boldsymbol{\beta}})^{\frac{-1}{2}}\exp\left[-\frac{1}{2}(\overline{\boldsymbol{\beta}}+\Sigma_{\boldsymbol{\beta}}^{1/2}Z_{-1}-\overline{\boldsymbol{\beta}})^T\Sigma_{\boldsymbol{\beta}}^{-1}(\overline{\boldsymbol{\beta}}+\Sigma_{\boldsymbol{\beta}}^{1/2}Z_{-1}-\overline{\boldsymbol{\beta}})\right]|\det(\Sigma_{\boldsymbol{\beta}}^{1/2})|d\boldsymbol{Z}\\ =&(2\pi)^{\frac{-K}{2}}\exp\left[-\frac{1}{2}(Z_{-1}^T(\Sigma_{\boldsymbol{\beta}}^{1/2})^T)\Sigma_{\boldsymbol{\beta}}^{-1}((\Sigma_{\boldsymbol{\beta}}^{1/2})^TZ_{-1})\right]d\boldsymbol{Z}\\ =&(2\pi)^{\frac{-K}{2}}\exp\left[-\frac{1}{2}Z_{-1}^TZ_{-1})\right]d\boldsymbol{Z}\end{aligned}$$

Therefore, given μ, the logit choice probability is bounded from above by 1 for all $(\mathbf{p},\Sigma)$. By (26), $\boldsymbol{\theta}$ is continuous in $\boldsymbol{Z}$, and the logit choice probabilities are continuous in prices and random coefficients. Therefore, by the Dominated Convergence Theorem, we can exchange limit and integrals. Consider any $(\mathbf{p},\Sigma)\in R_+^L\times S$ and any sequence $(\mathbf{p}_n,\Sigma_n)$ converging to it. If $\Sigma=\mathbf{0}_\Sigma$, under our change of variables the integral is well defined and equal to $s_l^{Lo}(\mathbf{p})$ for all l, for all $\mathbf{p}$. We know that

$$\lim_{n\to\infty} s_l^{Lo}(\mathbf{p}_n;\mu,\Sigma_n) \to s_l^{Lo}(\mathbf{p};\mu,\Sigma)$$

as $(\mathbf{p}_n\Sigma_n) \to (\mathbf{p}, \Sigma)$. Using the DCT we have

$$\begin{aligned}
\lim_{n\to\infty} s_l(\mathbf{p}_n; \mu, \Sigma_n) &= \lim_{n\to\infty} \int s_l^{Lo}(\mathbf{p}_n; \mu, \Sigma_n)\phi(\boldsymbol{Z})d\boldsymbol{Z} \\
&= \int \lim_{n\to\infty} s_l^{Lo}(\mathbf{p}_n; \mu, \Sigma_n)\phi(\boldsymbol{Z})d\boldsymbol{Z} \\
&= s_l(\mathbf{p}; \mu, \Sigma).
\end{aligned}$$

The continuity of profit functions and their first partial derivatives depends on the continuity of market shares on $(\mathbf{p}, \Sigma)$, because markups are continuous functions of prices, and from (5) and (6) we can use α_i as a dominating function to apply the DCT.

For the arrays of second partial derivatives we additionally require boundedness of the profits under logit and mixed logit demand to obtain a dominating function using α_i and α_i^2. Lemma 2.1 of Morrow and Skerlos (2010a) (which requires Assumption A.3) and Assumption A.11 of Morrow and Skerlos (2010b), which we know hold, deliver this boundedness, completing the proof. ■

A.5 Proof of Lemma 3.3

The proof is constructive: I write the bound $\overline{\mathbf{p}}(\Sigma)$ of Lemma 3.1 explicitly and show that every constant entering it is bounded uniformly on S_R. Throughout, μ is fixed and suppressed, the dependence on Σ is explicit, $K \equiv \dim(X)$, and

$$V_l(\boldsymbol{\beta}_i) \equiv \mathbf{x}_l^T\boldsymbol{\beta}_i + \xi_l, \qquad Q_l(\boldsymbol{\beta}_i) \equiv \frac{\exp(V_l(\boldsymbol{\beta}_i))}{\exp(u_0) + \sum_{h\in\mathcal{L}} \exp(V_h(\boldsymbol{\beta}_i))} \in (0, 1),$$

so that Q_l is the choice probability evaluated at $\mathbf{p} = \mathbf{0}$ and is a function of $\boldsymbol{\beta}_i$ alone. Normalize $\xi_0 = 0$, which is without loss because logit choice probabilities depend on utilities only through differences; if the normalization is not imposed, every constant C_{lk} below is multiplied by $\exp(-2\xi_0)$.

Step 1: A markup bound at any SSP. Fix $\Sigma \in S_R$ and let $\mathbf{p}^*$ be any SSP. By (9) and (10), $\mathbf{p}^* - \boldsymbol{c} = \zeta(\mathbf{p}^*; \Sigma)$. Under Assumption 2.1, $\Gamma(\mathbf{p}; \Sigma)$ is symmetric, and H is symmetric by construction, so $H \odot \Gamma(\mathbf{p}; \Sigma)$ is symmetric and $\widetilde{\Omega}(\mathbf{p}; \Sigma) = \Lambda(\mathbf{p}; \Sigma)^{-1}(H \odot \Gamma(\mathbf{p}; \Sigma))$ by (12). Hence

$$\mathbf{p}^* - \boldsymbol{c} = \widetilde{\Omega}(\mathbf{p}^*; \Sigma)(\mathbf{p}^* - \boldsymbol{c}) - \Lambda(\mathbf{p}^*; \Sigma)^{-1}\mathbf{s}(\mathbf{p}^*; \Sigma). \tag{29}$$

The triangle inequality and (25) give

$$\|\mathbf{p}^* - \boldsymbol{c}\|_\infty \leq \|\widetilde{\Omega}(\mathbf{p}^*; \Sigma)(\mathbf{p}^* - \boldsymbol{c})\|_\infty + \frac{1}{\underline{\alpha}}, \tag{30}$$

and the second bound is free of Σ and of $\mathbf{p}$.

Step 2: Markups are strictly positive at any SSP. By Corollary 2.8 of Morrow and Skerlos (2010a), whose Assumptions 2.1–2.3 correspond to Assumptions A.1, A.2, A.3, and A.4 and are verified in the proofs of Lemmas 2.5 and 3.1, profit-optimal markups are strictly positive; since the BLP-markup map and ζ share fixed points, any SSP satisfies $\mathbf{p}^* > \boldsymbol{c}$, and hence $0 < p_k^* - c_k < p_k^*$ for every $k \in \mathcal{L}$. For logit, the same conclusion follows from Lemma 4.2 of Morrow and Skerlos (2010b).

Step 3: Explicit bound on first term of (30). Repeating the argument used to verify Assumption A.5 in the proof of Lemma 3.1, but retaining the constants, (24) reads

$$\widetilde{\omega}_{lk}(\mathbf{p};\Sigma) \leq C_{lk}(\Sigma)\exp(-\underline{\alpha}p_k), \qquad C_{lk}(\Sigma) \equiv \frac{\mathbb{E}_\Sigma\left[\exp\left(V_l(\boldsymbol{\beta}_i) + V_k(\boldsymbol{\beta}_i)\right)\right]}{\mathbb{E}_\Sigma\left[Q_l(\boldsymbol{\beta}_i)\right]}, \tag{31}$$

where the expectations are taken over $\boldsymbol{\beta}_i \sim N(\overline{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}})$ only; the integrals in α_i cancel exactly, by the independence in Assumption 2.3.3 and the inequality $\exp(-\alpha_i p_k) \leq \exp(-\underline{\alpha}p_k)$ of Assumption 2.2. By Step 2 and $\sup_{p\geq 0} p\exp(-\underline{\alpha}p) = 1/(e\underline{\alpha})$,

$$\|\widetilde{\Omega}(\mathbf{p}^*;\Sigma)(\mathbf{p}^* - \boldsymbol{c})\|_\infty \leq \max_{l\in\mathcal{L}} \sum_{k\in\mathcal{L}_{j(l)}} C_{lk}(\Sigma)\exp(-\underline{\alpha}p_k^*)(p_k^* - c_k) \leq \underbrace{\max_{l\in\mathcal{L}} \sum_{k\in\mathcal{L}_{j(l)}} \frac{C_{lk}(\Sigma)}{e\underline{\alpha}}}_{\equiv\mathcal{K}(\Sigma)}. \tag{32}$$

Combining (30) and (32), and fixing any $\eta > 0$, the vector with entries

$$\overline{p}_l(\Sigma) \equiv c_l + \mathcal{K}(\Sigma) + \frac{1}{\underline{\alpha}} + \eta \tag{33}$$

satisfies the conclusion of Lemma 3.1: every SSP lies in the interior of $[\boldsymbol{c}, \overline{\mathbf{p}}(\Sigma)]$. Equation (33) is taken as the definition of $\overline{\mathbf{p}}(\Sigma)$ from here on. It remains to show $\sup_{\Sigma\in S_R}\mathcal{K}(\Sigma) < \infty$, which by (32) reduces to bounding the numerator of $C_{lk}(\Sigma)$ from above and its denominator away from zero, uniformly on S_R.

Step 4: Uniform upper bound on numerator. Since Σ is block diagonal, $\Sigma_{\boldsymbol{\beta}}$ is a submatrix of Σ, so under either reading of the maximum norm (entrywise maximum, or maximum absolute row sum) $\|\Sigma\|_\infty \leq R$ implies $\max_{g,h}|(\Sigma_{\boldsymbol{\beta}})_{gh}| \leq R$. Writing $\boldsymbol{v}_{lk} \equiv \mathbf{x}_l + \mathbf{x}_k$, we have $V_l(\boldsymbol{\beta}_i) + V_k(\boldsymbol{\beta}_i) = \boldsymbol{v}_{lk}^T\boldsymbol{\beta}_i + \xi_l + \xi_k$, which is normal with mean $\boldsymbol{v}_{lk}^T\overline{\boldsymbol{\beta}} + \xi_l + \xi_k$ and variance $\boldsymbol{v}_{lk}^T\Sigma_{\boldsymbol{\beta}}\boldsymbol{v}_{lk}$. The moment generating function of the normal distribution therefore gives

$$\mathbb{E}_\Sigma\left[\exp\left(V_l + V_k\right)\right] = \exp\left(\xi_l + \xi_k + \boldsymbol{v}_{lk}^T\overline{\boldsymbol{\beta}} + \tfrac{1}{2}\boldsymbol{v}_{lk}^T\Sigma_{\boldsymbol{\beta}}\boldsymbol{v}_{lk}\right).$$

Since $\boldsymbol{v}_{lk}^T\Sigma_{\boldsymbol{\beta}}\boldsymbol{v}_{lk} \leq \|\boldsymbol{v}_{lk}\|_1^2\max_{g,h}|(\Sigma_{\boldsymbol{\beta}})_{gh}| \leq R\|\boldsymbol{v}_{lk}\|_1^2$,

$$\sup_{\Sigma\in S_R}\mathbb{E}_\Sigma\left[\exp\left(V_l + V_k\right)\right] \leq \exp\left(\xi_l + \xi_k + (\mathbf{x}_l + \mathbf{x}_k)^T\overline{\boldsymbol{\beta}} + \tfrac{R}{2}\|\mathbf{x}_l + \mathbf{x}_k\|_1^2\right) \equiv \overline{N}_{lk}(R) < \infty. \tag{34}$$

Step 5: Uniform lower bound on denominator. Since $\mathbb{E}_\Sigma[Q_l(\boldsymbol{\beta}_i)]$ enters the denominator of $C_{lk}(\Sigma)$ in (31), we require $\inf_{\Sigma\in S_R}\mathbb{E}_\Sigma[Q_l(\boldsymbol{\beta}_i)] > 0$. This does not follow from the pointwise positivity $Q_l > 0$: the infimum of an uncountable set of positive numbers may be zero. I therefore show the existence of a lower bound that depends only on R and on $(\overline{\boldsymbol{\beta}}, (\mathbf{x}_h)_h, (\xi_h)_h, K)$.

Fix $\tau > 0$ and let $A_\tau \equiv \{\boldsymbol{\beta}\in\mathbb{R}^K : \|\boldsymbol{\beta} - \overline{\boldsymbol{\beta}}\|_\infty \leq \tau\}$. On A_τ we have $|V_h(\boldsymbol{\beta}) - V_h(\overline{\boldsymbol{\beta}})| \leq \|\mathbf{x}_h\|_1\tau$ for every $h \in \mathcal{L}$, so

$$\boldsymbol{\beta} \in A_\tau \implies Q_l(\boldsymbol{\beta}) \geq \frac{\exp\left(\mathbf{x}_l^T\overline{\boldsymbol{\beta}} + \xi_l - \|\mathbf{x}_l\|_1\tau\right)}{\exp(u_0) + \sum_{h\in\mathcal{L}} \exp\left(\mathbf{x}_h^T\overline{\boldsymbol{\beta}} + \xi_h + \|\mathbf{x}_h\|_1\tau\right)} \equiv q_l(\tau) > 0. \tag{35}$$

It remains to bound $\rho_\Sigma(A_\tau)$ below, uniformly on S_R. Because the maximum of finitely many terms exceeds τ if and only if one of them does,

$$A_\tau^c = \bigcup_{g=1}^{K} \left\{\boldsymbol{\beta} : |\beta_g - \overline{\beta}_g| > \tau\right\}.$$

Apply Chebyshev's inequality to each coordinate separately: for any $\Sigma \in S_R$ and any $g \in \{1, \ldots, K\}$, β_{ig} has mean $\overline{\beta}_g$ and variance $(\Sigma_{\boldsymbol{\beta}})_{gg} \leq R$ by Step 4, so $\rho_\Sigma\left(|\beta_{ig} - \overline{\beta}_g| > \tau\right) \leq (\Sigma_{\boldsymbol{\beta}})_{gg}/\tau^2 \leq R/\tau^2$. Countable subadditivity of ρ_Σ over the K coordinates then gives

$$\rho_\Sigma\left(A_\tau^c\right) \leq \frac{KR}{\tau^2} \qquad \forall\, \Sigma \in S_R. \tag{36}$$

Setting $\tau_R \equiv \sqrt{2KR}$ yields $\rho_\Sigma(A_{\tau_R}) \geq 1/2$ for every $\Sigma \in S_R$; this includes $\Sigma = \mathbf{0}_\Sigma$, at which $\boldsymbol{\beta}_i = \overline{\boldsymbol{\beta}}$ ρ-a.s. and $\rho_{\mathbf{0}_\Sigma}(A_{\tau_R}) = 1$. Combining (35) and (36) with $Q_l > 0$,

$$\inf_{\Sigma\in S_R} \mathbb{E}_\Sigma\left[Q_l(\boldsymbol{\beta}_i)\right] \geq \inf_{\Sigma\in S_R} q_l(\tau_R)\, \rho_\Sigma\left(A_{\tau_R}\right) \geq \tfrac{1}{2}\, q_l\left(\sqrt{2KR}\right) > 0. \tag{37}$$

Note: Since the α_i-integrals cancel in (31), all expectations below are with respect to the marginal of $\rho(\cdot; \mu, \Sigma)$ on $\boldsymbol{\beta}_i$, which by Assumption 2.3.3 is $N(\overline{\boldsymbol{\beta}}, \Sigma_{\boldsymbol{\beta}})$. I write ρ_Σ and $\mathbb{E}_\Sigma$ for that marginal and its expectation, leaving the dependence on the fixed μ implicit.

Conclusion. Combining (31), (34), and (37),

$$\sup_{\Sigma\in S_R} C_{lk}(\Sigma) \leq \frac{2\,\overline{N}_{lk}(R)}{q_l\left(\sqrt{2KR}\right)} \equiv \overline{C}_{lk}(R) < \infty \qquad \forall\, l, k \in \mathcal{L}.$$

So by (32) and (33),

$$\overline{p}_l(R) \equiv c_l + \max_{h\in\mathcal{L}} \sum_{k\in\mathcal{L}_{j(h)}} \frac{\overline{C}_{hk}(R)}{e\underline{\alpha}} + \frac{1}{\underline{\alpha}} + \eta$$

satisfies $\infty > \overline{p}_l(R) \geq \overline{p}_l(\Sigma) > c_l$ for every $\Sigma \in S_R$ and every $l \in \mathcal{L}$. Since $\mathbf{0}_\Sigma \in S_R$, the bound covers the logit model of Assumption 2.4. By Step 2 and (33), every SSP of (7) generated by any $\Sigma \in S_R$ lies in the interior of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. ■

Comment. Given R, the choice of a small η to construct $\overline{\mathbf{p}}(R)$ is desirable. This is because it reduces the search space for $\underline{d}$ and C_L described in Section 5.

A.6 Proof of Lemma 3.4

If S_R is compact for all R, uniform continuity follows from the compactness of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ and the Heine-Cantor Theorem: a continuous function between two metric spaces is uniformly continuous if its domain is compact.

To prove compactness of S_R we use the Heine-Borel Theorem on $\mathbb{R}^{\dim(\Theta)\times\dim(\Theta)}$. Boundedness of S_R in $\mathbb{R}^{\dim(\Theta)\times\dim(\Theta)}$ holds by its definition; it remains to prove the set is closed.

Fix $S = R > 0$ and pick any sequence of elements $A_n \in S_R$ such that $A_n \to A \in \mathbb{R}^{\dim(\Theta)\times\dim(\Theta)}$. We want to prove that $A \in S_R$. Letting i denote columns and j rows, using the maximum norm $A_n \to A$ implies that

$$\max_{ij} |a_{ij}^n - a_{ij}| \to 0 \quad \text{as} \quad n \to \infty \tag{38}$$

for all ij. If $i = 1$ and $j > 1$, then $a_{ij} = 0$ (and the same when $j = 1$ and $i > 1$). For all i. (38) implies that $a_{ii} \geq 0$ (nonnegative entries in the main diagonal). It also implies that, if $\mathbf{x} \in \mathbb{R}^{\dim(\Theta)}$ and $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^T A\mathbf{x} \geq 0$ (positive semi-definiteness). And it also implies that $a_{ij} = a_{ji}$ for all i, j (symmetry).

The constraint that every element of S be a variance covariance matrix means that $a_{ii} + a_{jj} \geq |2a_{ij}|$ for all i, j. Finally, because S_R is defined by a weak inequality and all $A^n] \in S_R$, it must be true that $\|A\|_\infty \leq R$. But then A satisfies all the conditions for being an element of S_R, so S_R is closed. This completes the proof. ■

A.7 Proof of Lemma 3.5

Preliminary 1: Choosing the space of functions. Since $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$ is bounded (Lemma 3.3) and the profits and their derivative are continuous (Lemma 2.5), the profit functions and their derivatives, constrained to a subset of their domain $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$, are bounded. Fix Σ (to index the functions) and consider profits and their derivatives as functions of prices from $E \equiv [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \to \mathbb{R}$. Endow E with the sup norm $\|\cdot\|_E$, so the distance between two functions in E is

$$\|f - g\|_E = \sup_{x\in E} |f(x) - g(x)|.$$

The space of functions we will consider is the space of bounded real functions of domain E, $\mathcal{B}(E, \mathbb{R})$, endowed with norm $\|f - g\|_E$. Note that because profits and their derivatives are defined over $\mathbb{R}^L_+ \times S$, the derivatives still exist on the boundary of E and, considering the constrained domain $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$, the functions are uniformly continuous in $(\mathbf{p}, \Sigma)$.

Preliminary 2: Distance function on $\mathbb{R}^L_+ \times S$ (and, for all $R > 0$. $E \times S_R$). We use the following distance function $d : \mathbb{R}^L_+ \times S \to \mathbb{R}$ (or $d_{|E\times S_R}$ when appropriate):

$$d((\mathbf{p}, \Sigma), (\mathbf{p}', \Sigma')) = |\mathbf{p} - \mathbf{p}'| + \|\Sigma - \Sigma'\|_\infty.$$

Preliminary 3: Pointwise convergence. Pick any $\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. Each profit function (and partial derivative functions) is indexed by a Σ_n. By the continuity of profit functions in $(\mathbf{p}, \Sigma)$, for each $j \in \mathcal{J}$ $\pi_j(\mathbf{p}; \mu, \Sigma_n) \to \pi_j(\mathbf{p}; \mu, \mathbf{0}_\Sigma)$ as $\Sigma_n \to \mathbf{0}_\Sigma$. The same is true for all entries of the arrays of first and second partial derivatives. This is pointwise convergence. Finally, note that each entry in the arrays of first and second partial derivatives is a real-valued function.

Part 1: Convergence of profits. Consider any sequence of Σ_n elements of $S_R \cap S_+$ that converges to $\mathbf{0}_\Sigma$. For all j, we want to prove that

$$\forall \varepsilon > 0,\, \exists N = N_\varepsilon \, s.t. \, \forall \mathbf{p} \in E,\, n \geq N_\varepsilon \Rightarrow \sup_{\mathbf{p}\in E} |\pi_j(\mathbf{p}; \Sigma_n) - \pi_j(\mathbf{p}; \mathbf{0}_\Sigma)| < \varepsilon.$$

By uniform continuity we have that, for all $\epsilon > 0$, there exists $\delta_\epsilon > 0$ such that, for all $(\mathbf{p}, \Sigma)$, $(\mathbf{p}', \Sigma') \in E \times S_R$, if $d((\mathbf{p}, \Sigma), (\mathbf{p}', \Sigma')) < \delta_\epsilon$ then $|\pi_j(\mathbf{p}, \Sigma) - \pi_j(\mathbf{p}', \Sigma')| < \epsilon$. Because $\Sigma_n \to \mathbf{0}_\Sigma$, for all $\delta > 0$ there exists N s.t. for all $n \geq N$, $\|\Sigma_n\|_\infty < \delta$. Now fix $\mathbf{p}$, so

$$d((\mathbf{p}, \Sigma_n), (\mathbf{p}, \mathbf{0}_\Sigma)) = \|\Sigma_n\|_\infty < \delta$$

implying, by uniform continuity,

$$|\pi_j(\mathbf{p}, \Sigma_n) - \pi_j(\mathbf{p}, \mathbf{0}_\Sigma)| < \epsilon$$

for all $\mathbf{p} \in E$. Taking the sup over E we have

$$\sup_{\mathbf{p}\in E} |\pi_j(\mathbf{p}, \Sigma_n) - \pi_j(\mathbf{p}, \mathbf{0}_\Sigma)| \leq \epsilon$$

yielding uniform convergence of profits.

Part 2: Convergence of partial derivatives. The argument of part 1 remains the same if we substitute the first (resp. second) partial derivative of any profit function w.r.t. any p_l, $l \in \mathcal{L}$ (resp. any p_l, p_k, l, $k \in \mathcal{L}$) because they are all uniformly continuous in $E \times S_R$. Indeed, since there are finitely many firms, products, and derivatives, we may choose a common $\varepsilon > 0$ for all of them. ■

A.8 Proof of Lemma 4.1

We use Kellogg (1976) and Konovalov and Sándor (2010). The first-order condition system is (7) and its Jacobian is (8), so we have already proven they are both continuous in $(\mathbf{p}, \Sigma) \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$ for all $R > 0$. By the composition of continuous functions, the determinant of (8), denoted here as $\det(DJ(\mathbf{p}, \Sigma))$ to simplify notation, is also continuous in $(\mathbf{p}, \Sigma) \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$ for all $R > 0$.

Denote $D_{\mathbf{p}} \left([D_{\mathbf{p}_j} \pi_j(\mathbf{p}_j, \mathbf{p}_{-j}; \mu, \Sigma)]_j \right)$ as $DF(\mathbf{p}, \Sigma)$ for simplicity. From Lemma 3.5 we have

$$\sup_{\mathbf{p}\in[\boldsymbol{c},\overline{\mathbf{p}}(R)]} |DF(\mathbf{p}, \Sigma_n) - DF(\mathbf{p}, \mathbf{0}_\Sigma)| \to 0$$

as $\Sigma_n \to \mathbf{0}_\Sigma$ (uniform convergence). This means that, given $Q > 0$, for large enough n, all entries of $DF(\mathbf{p}, \Sigma_n)$ have absolute value Q at most. By the definition of the determinant of a function $A \in \mathbb{R}^{L\times L}$ of entries $a_{i,j}$

$$\det(A) = \sum_{\tau\in T_L} \text{sign}(\tau) \prod_{i=1}^{L} a_{i,\tau(i)}$$

where τ is a permutation and T_L contains the $L!$ permutations of $\{1, ..., L\}$, we have that

$$\det(A) - \det(B) = \sum_{\tau \in T_L} \operatorname{sign}(\tau) \left[\prod_{i=1}^{L} a_{i,\tau(i)} - \prod_{i=1}^{L} b_{i,\tau(i)} \right]$$

Within each permutation we can rewrite the difference of products as

$$\prod_{i=1}^{L} a_{i,\tau(i)} - \prod_{i=1}^{L} b_{i,\tau(i)} = \sum_{l=1}^{L} \left(\prod_{i<l}^{L} a_{i,\tau(i)} \right) (a_{l,\tau(l)} - b_{l,\tau(l)}) \left(\prod_{i>l}^{L} b_{i,\tau(i)} \right)$$

where the intermediate terms will cancel out, as in a telescoping series. This equality is useful because if all entries of the matrices have absolute value of at most Q, we can bound

$$\prod_{i=1}^{L} a_{i,\tau(i)} - \prod_{i=1}^{L} b_{i,\tau(i)} \leq \sum_{l=1}^{L} (2Q)^{L-1} |(a_{l,\tau(l)} - b_{l,\tau(l)})| \leq L(2Q)^{L-1} \max_{l} |(a_{l,\tau(l)} - b_{l,\tau(l)})|.$$

Because there are $L!$ permutations we get

$$\det(A) - \det(B) \leq L! L(2Q)^{L-1} \max_{i,j} |a_{i,j} - b_{i,j}|.$$

Note then that for large enough N, there exists a finite $Q > 0$ such that

$$|\det(DF(\mathbf{p}, \Sigma_n)) - \det(DF(\mathbf{p}, \mathbf{0}_\Sigma))| \leq L! L(2Q)^{L-1} \max_{i,j} |DF_{i,j}(\mathbf{p}, \Sigma_n) - DF_{i,j}(\mathbf{p}, \mathbf{0}_\Sigma)|$$

where i, j denotes the derivative of the i-th entry of (7) with respect to the price of good j. By Lemma 3.5 we have

$$\sup_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} |\det(DF(\mathbf{p}, \Sigma_n)) - \det(DF(\mathbf{p}, \mathbf{0}_\Sigma))| \leq L! L(2Q)^{L-1} \sup_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} \max_{i,j} |DF_{i,j}(\mathbf{p}, \Sigma_n) - DF_{i,j}(\mathbf{p}, \mathbf{0}_\Sigma)|$$

where the right-hand side goes to zero as $\Sigma_n \to \mathbf{0}_\Sigma$. Since $\det(DF(\mathbf{p}, \Sigma))$ is real valued and bounded in $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$, we get uniform convergence of the determinant in the same space of funntions we used for the proof of Lemma 3.5.

Because $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ is compact, by Lemma 5 in Konovalov and Sándor (2010) and the extreme value theorem we have

$$\min_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} \det(DJ(\mathbf{p}, \mathbf{0}_\Sigma)) = \underline{d} > 0.$$

Therefore, if

$$\sup_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} |\det(DF(\mathbf{p}, \Sigma_n)) - \det(DF(\mathbf{p}, \mathbf{0}_\Sigma))| \to 0$$

as $\Sigma_n \to \mathbf{0}_\Sigma$, we have that there exists N such that

$$\sup_{\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]} \det(DF(\mathbf{p}, \Sigma_n)) > 0 \qquad \forall n \geq N$$

Therefore, there exists some $\varepsilon_1 > 0$ such that for all $\Sigma \in B_{S_R}(\mathbf{0}_\Sigma, \varepsilon_1) \backslash \{\mathbf{0}_\Sigma\}$, $\det(DF(\mathbf{p}, \Sigma)) > 0$ for all

$\mathbf{p} \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. Then, by Kellogg's theorem, (7) has a unique fixed point. ■

A.9 Proof of Lemma 4.2

In this proof I consider only $\Sigma \in B(\mathbf{0}_\Sigma, \varepsilon_1)$, so that (7) has a unique SSP, denoted $\mathbf{p}^*(\Sigma)$. The point of departure is Morrow and Skerlos (2010b)'s proof that, under logit demand, if Assumptions A.8 through A.11 hold, and if a firm's FOC hold at $\mathbf{p}_j$ given $\mathbf{p}_{-j}$, then the firm's profits are locally concave, so $\mathbf{p}_j$ is a local maximizer given $\mathbf{p}_{-j}$. The prove this by showing that if $\mathbf{p}_j$ satisfies j's FOCs given $\mathbf{p}_{-j}$, then the Hessian is a diagonal matrix with negative entries in the diagonal, so it is negative definite. By the proofs of Lemmas 2.5 and 3.1 we know that Assumption 2.4 implies Assumptions A.8 through A.11 hold.

Consider any sequence Σ_n of elements of $B(\mathbf{0}_\Sigma, \varepsilon_1)$. Then, the sequence $\mathbf{p}^*(\Sigma_n)$ is well defined. Claim A.12 presents a useful convergence result for our proof.

Claim A.12. *Let the conditions of Lemma 4.2 hold and let every term of Σ_n be an element of $B(\mathbf{0}_\Sigma, \varepsilon_1)$. If $\Sigma_n \to \mathbf{0}_\Sigma$, then $\mathbf{p}^*(\Sigma_n) \to \mathbf{p}^*(\mathbf{0}_\Sigma)$.*

Proof of Claim A.12. Morrow and Skerlos (2010b) present the logit equivalent of the ζ-markup (9) and the ζ-vector field (10) for logit. Due to uniform convergence, the ζ-vector field for mixed logit $(F_\zeta(\mathbf{p}; \Sigma))$ converges uniformly to the ζ-vector field for logit $(F_\zeta(\mathbf{p}; \mathbf{0}_\Sigma))$. Using the SSP condition we have

$$F_\zeta(\mathbf{p}^*(\Sigma); \Sigma) = 0 \iff \mathbf{p}^*(\Sigma) = \boldsymbol{c} + \zeta(\mathbf{p}^*(\Sigma); \Sigma)$$

so

$$\|\mathbf{p}^*(\mathbf{0}_\Sigma) - \mathbf{p}^*(\Sigma_n)\| = \|\zeta(\mathbf{p}^*(\mathbf{0}_\Sigma); \mathbf{0}_\Sigma) - \zeta(\mathbf{p}^*(\Sigma_n); \Sigma_n)\|.$$

But $\zeta(\cdot, \cdot)$ converges uniformly to $\zeta(\cdot, \mathbf{0}_\Sigma)$, yielding the proof. ■

From Lemma 3.5, as $\Sigma_n \to \mathbf{0}_\Sigma$, the Hessian $D^2_{\mathbf{p}_j,\mathbf{p}_j}\pi_j(\mathbf{p}^*(\Sigma_n); \Sigma_n)$ of each $j \in \mathcal{J}$ converges uniformly to $D^2_{\mathbf{p}_j,\mathbf{p}_j}\pi_j(\mathbf{p}^*(\mathbf{0}_\Sigma); \mathbf{0}_\Sigma)$, which is diagonal with negative entries in the diagonal. To simplify notation, ignore the subscript j and denote $D^2_{\mathbf{p}_j,\mathbf{p}_j}\pi_j(\mathbf{p}^*(\Sigma_n); \Sigma_n)$ as H^n and $D^2_{\mathbf{p}_j,\mathbf{p}_j}\pi_j(\mathbf{p}^*(\mathbf{0}_\Sigma); \mathbf{0}_\Sigma)$ as H.

Fixing any j and row $r \in \{1, ..., |\mathcal{L}_j|\}$ of $H(\mathbf{p})$. Then, $H_{rr} < 0$ and $H_{rs} = 0$ for all $s \neq r$, where s indexes the column. By uniform convergence, there exists $\delta_{jr} \in (0, |H_{rr}|/2)$, and some $N(\delta_{jr}) \in \mathbb{N}$, such that, for all $n \geq N(\delta_{jr})$, $|H^n_{rs}| < \delta_{jr}/(|\mathcal{L}_j| + 1)$ and $|H_{rr} - H^n_{rr}| < \delta_{jr}$. Therefore, an upper bound on H^n_{rr} is $H_{rr} + \delta_{jr} < 0$, with absolute value $|H_{rr} + \delta_{jr}| = -H_{rr} - \delta_{jr}$. The sum of the absolute values of the off-diagonal terms is

$$\sum_{s \neq r} |H^n_{rs}| \leq \frac{\delta_{jr}}{|\mathcal{L}_j| + 1} = \delta_{jr}\frac{|\mathcal{L}_j| - 1}{|\mathcal{L}_j| + 1} \in [0, \delta_{jr}).$$

Therefore $|H^n_{rr}| \geq -H_{rr} - \delta_{jr}$,and

$$\begin{aligned} |H^n_{rr}| - \sum_{s \neq r} |H^n_{rs}| &\geq -H_{rr} - \delta_{jr} - \delta_{jr}\frac{|\mathcal{L}_j| - 1}{|\mathcal{L}_j| + 1} \\ &\geq -H_{rr} - \delta_{jr} - \delta_{jr} > 0 \end{aligned}$$

because $\delta_{jr} < |H_{rr}|/2$ by Assumption. Since the number of rows is finite, given j define

$$N_j = \max_{r \in \{1,\ldots,|\mathcal{L}_j|\}} N(\delta_{jr}).$$

Because there is a finite number of firms, we can define a finite $\overline{N} = \max_j N_j$. Therefore, for all j, if $n \geq \overline{N}$, then $D^2_{\mathbf{p}_j,\mathbf{p}_j} \pi_j(\mathbf{p}^*(\Sigma_n); \Sigma_n)$ is diagonally dominant, meaning j's profit function is locally concave at the unique SSP of (7). Therefore, there exists $\varepsilon_2 > 0$ such that, for all $\Sigma \in B(\mathbf{0}_\Sigma, \varepsilon_2) \backslash \{\mathbf{0}_\Sigma\}$, where $B(\mathbf{0}_\Sigma, \varepsilon_2)$ is a nonempty open ball in S_R, the mixed logit model indexed by Σ has a unique SSP for (7), and at that SSP $\mathbf{p}^*(\Sigma)$ all firms' profits are locally concave and locally maximized with respect to $\mathbf{p}_j$ given $\mathbf{p}^*_{-j}(\Sigma)$.

Local concavity implies that for all $j \in \mathcal{J}$, j's profits are locally maximized at $\mathbf{p}^*_j(\Sigma)$ given $\mathbf{p}^*_{-j}(\Sigma)$, for Σ close enough to $\mathbf{0}_\Sigma$. Now I follow the steps in the second part of the proof of Lemma 4.12 of Morrow and Skerlos (2010b) and prove that $\mathbf{p}^*_j(\Sigma)$ is a global maximizer given $\mathbf{p}^*_{-j}(\Sigma)$ using results from Milnor (1965).

Let $\zeta_j(\cdot;\cdot)$ be the restriction of the vector valued ζ-markup equation to the entries associated with prices of firm j. For each j, the FOC $\mathbf{p}_j = \boldsymbol{c}_j + \zeta_j(\mathbf{p}; \Sigma)$ is a necessary condition for local maximization of $\pi_j(\cdot, \mathbf{p}_{-j}; \Sigma)$. By the proof of negative-definiteness above, we have, for Σ

$$\operatorname{sign}\det\left(D^2_{\mathbf{p}_j,\mathbf{p}_j} \pi_j(\mathbf{p}^*(\Sigma_n); \Sigma_n)\right) = (-1)^{L_j} \tag{39}$$

But by Lemma 4.1 of Morrow and Skerlos (2010a) (which requires no additional assumptions) we have

$$\begin{aligned}\operatorname{sign}\det\left(D^2_{\mathbf{p}_j,\mathbf{p}_j} \pi_j(\mathbf{p}^*(\Sigma_n); \Sigma_n)\right) &= \operatorname{sign}\det\left(\Lambda_j(\mathbf{p}; \Sigma) D_{\mathbf{p}_j} F_{\zeta,j}(\mathbf{p}; \Sigma)\right) \\ &= \underbrace{\operatorname{sign}\det\left(\Lambda_j(\mathbf{p}; \Sigma)\right)}_{=(-1)^{L_j}} \cdot \operatorname{sign}\det\left(D_{\mathbf{p}_j} F_{\zeta,j}(\mathbf{p}; \Sigma)\right). \end{aligned} \tag{40}$$

Putting (39) and (40) together we get

$$\operatorname{sign}\det\left(D_{\mathbf{p}_j} F_{\zeta,j}(\mathbf{p}; \Sigma)\right) = 1.$$

By the Poincaré-Hopf theorem (Milnor, 1965), if (1) $F_{\zeta,j}(\cdot, \Sigma)$ is continuously differentiable with respect to $\mathbf{p}$ (Lemma 2.5), (2) the Euler characteristic of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ is 1 (true by convexity of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$), and (3) $F_{\zeta,j}(\cdot, \Sigma)$ points outwards at the boundary of $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$ (Corollary 2.19 of Morrow and Skerlos (2010a), whose conditions hold under Assumptions 2.1 through 2.3), then the sum of indices at its roots is 1. But if all the roors have index 1, then there is only one root. Therefore, each firm has only one local maximizer when $\mathbf{p}_{-j} = \mathbf{p}^*_{-j}(\Sigma)$, and that vector is $\mathbf{p}^*_j(\Sigma)$. ■

A.10 Proof of Theorem 4.3

Follows from the composition of Lemmas 4.1 and 4.2. Trivially, $\emptyset \neq B(\mathbf{0}_\Sigma, \varepsilon_2) \backslash \{\mathbf{0}_\Sigma\} \subset B(\mathbf{0}_\Sigma, \varepsilon_1) \backslash \{\mathbf{0}_\Sigma\} \subset \mathcal{BN}$. $B(\mathbf{0}_\Sigma, \varepsilon_1) \backslash \{\mathbf{0}_\Sigma\} \neq \mathcal{BN} \backslash \{\mathbf{0}_\Sigma\}$ is possible, because the proofs of Lemmas 4.1 and 4.2 give sufficient but not necessary conditions for existence and uniqueness of a Bertrand-Nash equilibrium. ■

A.11 Proof of Lemma 5.1

Fix $\epsilon > 0$ and define the open, convex box

$$\mathcal{C}_R^\epsilon \equiv \prod_{l\in\mathcal{L}} (c_l - \epsilon,\ \overline{p}_l(R) + \epsilon) \supset [\boldsymbol{c}, \overline{\mathbf{p}}(R)], \tag{41}$$

and let $B_l(R,\epsilon) \equiv \overline{p}_l(R) - c_l + \epsilon$, so that $|p_l - c_l| \le B_l(R,\epsilon)$ for all $\mathbf{p} \in \mathcal{C}_R^\epsilon$. Set $A_0 \equiv 1$ and $A_n \equiv 2^n n!$ for $n \ge 1$. Throughout, $g_j(\mathbf{p},\boldsymbol{\theta}_i) \equiv \sum_{l\in\mathcal{L}_j}(p_l - c_l)s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)$ denotes the integrand of (4), so that $\pi_j(\mathbf{p};\mu,\Sigma) = \int g_j(\mathbf{p},\boldsymbol{\theta}_i)\, d\rho(\widetilde{\boldsymbol{\theta}}_i;\mu,\Sigma)$.

Step 1: Derivatives of logit choice probabilities. Under Assumption 2.1,

$$\frac{\partial s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)}{\partial p_m} = -\alpha_i\, s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)\left(\mathbf{1}[l=m] - s_m^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)\right). \tag{42}$$

Claim A.13. *For every $n \ge 1$ and every $(a_1,\dots,a_n) \in \mathcal{L}^n$,*

$$\frac{\partial^n s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)}{\partial p_{a_1}\cdots\partial p_{a_n}} = \alpha_i^n\, P^{(n)}\left(s_1^{Lo}(\mathbf{p},\boldsymbol{\theta}_i),\dots,s_L^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)\right), \tag{43}$$

where $P^{(n)}$ is a polynomial with integer coefficients, of total degree at most $n+1$, whose coefficients sum in absolute value to at most $A_n < \infty$.

Proof of Claim A.13. The proof is by induction. For $n=1$ this is (42), with $P^{(1)} = s_l s_m - \mathbf{1}[l=m]s_l$: degree 2 and absolute coefficient sum $2 = A_1$. Suppose the claim holds at n. For a monomial $\prod_{t=1}^d s_{k_t}^{Lo}$ of degree $d \le n+1$, (42) and the product rule give

$$\frac{\partial}{\partial p_m}\prod_{t=1}^d s_{k_t}^{Lo} = \alpha_i \sum_{t=1}^d \left[\Big(\prod_{r=1}^d s_{k_r}^{Lo}\Big)s_m^{Lo} - \mathbf{1}[k_t = m]\prod_{r=1}^d s_{k_r}^{Lo}\right],$$

a polynomial of degree at most $d+1$ whose absolute coefficient sum is at most $2d$ times that of the original monomial. Applying this termwise to $P^{(n)}$ yields $P^{(n+1)}$ of degree at most $n+2$ with absolute coefficient sum at most $2(n+1)A_n = A_{n+1}$, proving the claim. Since $s_k^{Lo}(\mathbf{p},\boldsymbol{\theta}_i) \in (0,1)$ for all k, all $\mathbf{p}$, and all $\boldsymbol{\theta}_i$, we conclude that for every $n \ge 1$, every multi-index, every $\mathbf{p} \in \mathbb{R}_+^L$ and ρ-a.e. $\boldsymbol{\theta}_i$,

$$\left|\frac{\partial^n s_l^{Lo}(\mathbf{p},\boldsymbol{\theta}_i)}{\partial p_{a_1}\cdots\partial p_{a_n}}\right| \le A_n\,\alpha_i^n. \tag{44}$$

Step 2: Dominating function on $\mathcal{C}_R^\epsilon$. Fix $n \in \{1,2,3\}$ and a multi-index $(a_1,\dots,a_n)$. Because $p_l - c_l$ is affine in $\mathbf{p}$, Leibniz's rule implies that at most one of the n derivatives can act on it:

$$\frac{\partial^n\left[(p_l - c_l)s_l^{Lo}\right]}{\partial p_{a_1}\cdots\partial p_{a_n}} = (p_l - c_l)\frac{\partial^n s_l^{Lo}}{\partial p_{a_1}\cdots\partial p_{a_n}} + \sum_{t=1}^n \mathbf{1}[a_t = l]\,\frac{\partial^{n-1}s_l^{Lo}}{\prod_{r\ne t}\partial p_{a_r}}.$$

Combining this with (44) and $|p_l - c_l| \le B_l(R,\epsilon)$ on $\mathcal{C}_R^\epsilon$,

$$\sup_{\mathbf{p}\in\mathcal{C}_R^\epsilon}\left|\frac{\partial^n g_j(\mathbf{p},\boldsymbol{\theta}_i)}{\partial p_{a_1}\cdots\partial p_{a_n}}\right| \;\le\; \sum_{l\in\mathcal{L}_j}\Big[B_l(R,\epsilon)\,A_n\,\alpha_i^n + n\,A_{n-1}\,\alpha_i^{n-1}\Big] \;\equiv\; G_n(\alpha_i). \tag{45}$$

Note that G_n depends on $\boldsymbol{\theta}_i$ only through α_i, and is free of $\mathbf{p}$ and of Σ. Note also that the term of order α_i^n is the highest that can appear: no term in α_i^{n+1} or higher arises.

Step 3: Integrability (uniformly in $\Sigma \in S_R$). Under (26), $\alpha_i(z_1,\Sigma) = \underline{\alpha} + \exp(\mu_\alpha + \sigma_\alpha z_1)$ with $z_1 \sim N(0,1)$ under a Σ-free law. Since $\sigma_\alpha^2 = \Sigma_{11}$ and $\|\Sigma\|_\infty \le R$, we have $\sigma_\alpha \in [0,\sqrt{R}]$. The map $\sigma_\alpha \mapsto \exp(\mu_\alpha + \sigma_\alpha z_1)$ is increasing for $z_1 \ge 0$ and decreasing for $z_1 < 0$, so

$$\sup_{\Sigma\in S_R}\alpha_i(z_1,\Sigma) = \underline{\alpha} + \exp\Big(\mu_\alpha + \sqrt{R}\,z_1^+\Big) \;\le\; \underline{\alpha} + \exp\Big(\mu_\alpha + \sqrt{R}\,|z_1|\Big) \;\equiv\; \overline{\alpha}_R(z_1), \tag{46}$$

a function of z_1 alone. Since $|z_1|$ is half-normal, completing the square gives, for any $a \ge 0$, $\mathbb{E}[e^{a|z_1|}] = 2e^{a^2/2}\Phi(a) \le 2e^{a^2/2}$, where Φ is the cdf of the standard normal distribution function.[8] Expanding $\overline{\alpha}_R^n$ by the binomial theorem and applying this bound termwise at $a = k\sqrt{R}$,

$$\mathbb{E}\left[\overline{\alpha}_R(z_1)^n\right] \;\le\; 2\sum_{k=0}^n \binom{n}{k}\underline{\alpha}^{n-k}\exp\left(k\mu_\alpha + \frac{k^2 R}{2}\right) \;<\; \infty \qquad \text{for every } n \ge 0. \tag{47}$$

Step 4: Differentiation under the integral sign. Fix $\Sigma \in S_R$. I show by induction on $n \in \{1,2,3\}$ that $\pi_j(\cdot;\mu,\Sigma)$ is n times differentiable on $\mathcal{C}_R^\epsilon$ with

$$\frac{\partial^n \pi_j(\mathbf{p};\mu,\Sigma)}{\partial p_{a_1}\cdots\partial p_{a_n}} = \int \frac{\partial^n g_j(\mathbf{p},\boldsymbol{\theta}_i)}{\partial p_{a_1}\cdots\partial p_{a_n}}\,d\rho(\widetilde{\boldsymbol{\theta}}_i;\mu,\Sigma). \tag{48}$$

Suppose (48) holds at order $n-1$ (the case $n=1$ being (4) itself). The integrand $\partial^{n-1}g_j(\cdot,\boldsymbol{\theta}_i)$ is differentiable on $\mathcal{C}_R^\epsilon$ for ρ-a.e. $\boldsymbol{\theta}_i$ by Step 1. For $\mathbf{p} \in \mathcal{C}_R^\epsilon$, $t \neq 0$ small enough that the segment $[\mathbf{p}, \mathbf{p} + t\boldsymbol{e}_{a_n}]$ lies in $\mathcal{C}_R^\epsilon$, convexity of $\mathcal{C}_R^\epsilon$ and the mean value theorem give

$$\left|\frac{\partial^{n-1}g_j(\mathbf{p}+t\boldsymbol{e}_{a_n},\boldsymbol{\theta}_i) - \partial^{n-1}g_j(\mathbf{p},\boldsymbol{\theta}_i)}{t}\right| \le \sup_{\mathbf{p}'\in\mathcal{C}_R^\epsilon}\left|\frac{\partial^n g_j(\mathbf{p}',\boldsymbol{\theta}_i)}{\partial p_{a_1}\cdots\partial p_{a_n}}\right| \le G_n(\alpha_i),$$

which is ρ-integrable by Step 3. By Theorem 2.27 of **?**, (48) holds at order n. Taking $n = 3$ establishes existence of all third partial derivatives of π_j on $\mathcal{C}_R^\epsilon$, and hence on the open set $\operatorname{int}[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$.

Step 5: Joint continuity of third derivatives. Let $(\mathbf{p}_h, \Sigma_h) \to (\mathbf{p}, \Sigma)$ in $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$. Under (26) the principal matrix square root $\Sigma \mapsto \Sigma_{\boldsymbol{\beta}^{1/2}}$ is continuous on the positive semidefinite cone, so $\Sigma \mapsto \boldsymbol{\theta}_i(\boldsymbol{Z},\Sigma)$ is continuous for every $\boldsymbol{Z}$; and by Step 1 each third partial of g_j is a continuous function of $(\mathbf{p},\boldsymbol{\theta}_i)$. Hence $\partial^3 g_j(\mathbf{p}_h, \boldsymbol{\theta}_i(\boldsymbol{Z},\Sigma_h)) \to \partial^3 g_j(\mathbf{p}, \boldsymbol{\theta}_i(\boldsymbol{Z},\Sigma))$ pointwise in $\boldsymbol{Z}$. By (45) and (46) the sequence is dominated by $\overline{G}_3(z_1) \in L^1$, which does not depend on h. The dominated convergence theorem then gives, using

[8] For $z \sim N(0,1)$ and $a \ge 0$, symmetry gives $\mathbb{E}[e^{a|z|}] = 2\int_0^\infty e^{az}\phi(z)dz$; since $az - z^2/2 = -(z-a)^2/2 + a^2/2$, this equals $2e^{a^2/2}\int_{-a}^\infty \phi(u)du = 2e^{a^2/2}\Phi(a)$.

(48),

$$\frac{\partial^3 \pi_j(\mathbf{p}_h; \mu, \Sigma_h)}{\partial p_a \partial p_b \partial p_d} \longrightarrow \frac{\partial^3 \pi_j(\mathbf{p}; \mu, \Sigma)}{\partial p_a \partial p_b \partial p_d}.$$

The same argument applied with G_1 and G_2 gives the analogous statement for the first and second partial derivatives. Therefore π_j and its first, second, and third partial derivatives with respect to $\mathbf{p}$ are jointly continuous on $[\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$; in particular, for each fixed $\Sigma \in S_R$, $\pi_j(\cdot; \mu, \Sigma)$ is three times continuously differentiable on $\operatorname{int}[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$, with third derivatives extending continuously to $[\boldsymbol{c}, \overline{\mathbf{p}}(R)]$. ■

Comment. Steps 1 through 5 are stated for a general order n and used only at $n \leq 3$. Since $\mathbb{E}_\Sigma[\overline{\alpha}_R(z_1)^n] < \infty$ for every n (Assumption 2.3.1), the same induction delivers continuous differentiability of any order q of mixed logit profits with respect to $\mathbf{p}$ on $\mathcal{C}_R^\epsilon$, jointly continuous in $(\mathbf{p}, \Sigma) \in [\boldsymbol{c}, \overline{\mathbf{p}}(R)] \times S_R$.

B Uniform Convergence

This appendix presents a more general proof of the uniform convergence result, which may be applicable to multiple economic models.

Let X and Y be nonempty, compact subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Let $C(X \times Y, \mathbb{R})$ be the set of continuous functions from $X \times Y$ to $\mathbb{R}$. Because X and Y are compact, these functions are bounded and, by the Heine-Cantor Theorem, also uniformly continuous.

Fixing any $y \in Y$, the fiber $f(\cdot, y)$ of any element $f \in C(X \times Y, \mathbb{R})$ is an element of $C(X, \mathbb{R})$ (real valued, bounded, uniformly continuous). We will consider sequences $f_n \in C(X \times Y, \mathbb{R})$, where each element of the sequence is a fiber of the same function $f \in C(X \times Y, \mathbb{R})$, indexed by some $y \in Y$.

Lemma B.1. *Let $f \in C(X \times Y, \mathbb{R})$. Let the sequence of functions f_n elements of $C(X, \mathbb{R})$ be defined as fibers of f, indexed by (possibly different) elements $y \in Y$. We can therefore write f_n as $f(\cdot, y_n)$, $y_n \in Y$. Let $y_0 \in Y$ and let $y_n \to y_0$. hen, $f(\cdot, y_n)$ converges uniformly to $f(\cdot, y_0)$.*

Proof. Begin with pointwise convergence. Fixing $x \in X$. $f(x, y_n) \to f(x, y_0)$ by the continuity of f in $X \times Y$.

To prove uniform convergence, we want to prove that, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for all $n > N$, for all $x \in X$, $|f(x, y_n) - f(x, y_0)| < \varepsilon$. Alternatively, we can write the last condition as

$$\sup_{x \in X} |f(x, y_n) - f(x, y_0)| \to 0 \quad \text{as} \quad n \to \infty.$$

By uniform continuity, let $(a, b) \in X \times Y$. Then, for all $\varepsilon > 0$, $\exists \delta > 0$ s.t. $\|x - a\| + \|y - b\| < \delta$ implies $|f(x, y) - f(a, b)| < \varepsilon$ (because X and Y are finite dimensional the choice of norm does not affect this result).

Fix $\varepsilon > 0$. Then, there exists $\delta > 0$ s.t., for all $x \in X$, $\|y - y_0\| < \delta$ implies $|f(x, y_n) - f(x, y_0)| < \varepsilon$. Since $y_n \to y_0$, there exists N large enough that $\|y_n - y_0\|$ for all $n > N$. Then, there exists N such that, for all $n > N$,

$$\sup_{x \in X} |f(x, y_n) - f(x, y_0)| < \varepsilon$$

for all $\varepsilon > 0$, completing the proof. ■

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